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Inverted Assembly of the Lens Within Ocular Organoids Reveals Alternate Paths to Ocular Morphogenesis

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eLife Assessment

This **important** study demonstrates that ocular organoids can generate both retina and lens through a non-canonical, "inside-out" morphogenetic route. The work is **solid**, with well-designed experiments combining imaging, molecular analyses, and transcriptomics to establish that lens formation in organoids follows conserved molecular programs despite an alternative morphogenesis. These findings expand our understanding of self-organization and developmental plasticity, and will be of broad interest to researchers working on eye development, organoids, and tissue engineering.

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Abstract

The eye is a complex organ composed of two main structures – the retina and the lens. It forms by the invagination of the lens forming head surface ectoderm embedding into the forming optic cup. This “outside-in” mode of morphogenesis ensures that the light focusing lens is positioned centrally inside of the eye in the highly constrained environment of the developing embryo. Advances in stem cell biology in the last decade introduced organoids as model to study organogenesis under normal and diseased conditions. However, even though strikingly similar at some points, it remained elusive to which extend the generation of individual structural features in organoids recapitulates *in vivo* organogenesis. Here we describe the generation of fish ocular organoids composed of both, lens and retina, using pluripotent embryonic cells from medaka (*Oryzias latipes*). Formation of the organoid lens followed the key molecular features of the process *in vivo*, including the establishment of lens progenitor cells and their subsequent differentiation into lens fiber cells. In a process dependent on the coordinated activity of BMP and FGF signaling, lens formation in ocular organoids was marked by the expression of key genes implicated in organismal lens development. Despite adhering to the basic molecular machinery of lens formation *in vivo*, the morphogenesis into a spherical lens followed an “inside-out” mode. Lens progenitor cells were initially established and differentiated into a spherical lens directly inside of the retina.

Subsequent displacement of the lens from the center of the organoid towards its surface ultimately led to the formation of a cup-shaped retina with a centrally positioned lens. Our study highlights that the self-organization of the organoid can favor routes that were not selected for in the developing embryo. Those routes can lead to an alternative, though highly similar outcome with the respect to achieving specific structural features in an unconstrained, embryo-free environment.

Introduction

The tissues of the embryo develop in an evolutionarily selected confined space with robust physical and biochemical constraints. Precise spatio-temporal gradients of growth factors aligned with complex mechanical forces guide cell and tissue interactions, permitting the coordinated formation of functional tissues and organs. Eye development is an example of complex organ formation relying on the coordinated interactions of tissues of different embryonic origin. This ultimately results in the positioning of a light-focusing lens in the center of the light-sensing retina (Cardozo et al., 2023; Casey et al., 2021; Chow and Lang, 2001; Cvekl and Ashery-Padan, 2014).

While the lens originates from the head surface ectoderm (SE), the retina originates from optic vesicle (OV), a lateral evagination of the forming diencephalon. In vertebrates, eye development is initiated at the end of gastrulation when the anterior ectoderm is patterned into the neural plate, non-neuronal ectoderm and the neural plate border at their interface. Lens-committed progenitors are first found in the anterior neural plate border, also called the anterior pre-placodal region (Bailey and Streit, 2005; Bhattacharyya et al., 2004; McCabe and Bronner-Fraser, 2009; Schlosser, 2006; Toro and Varga, 2007). Retina-committed progenitors, on the other hand, are established within the anterior neural plate, by the specification of the eye field, followed by the evagination of the optic vesicle (Adelmann, 1937; Eagleson et al., 1995; Kenyon et al., 2001; Li et al., 1997). Once the retina-forming OV contacts the lens-competent ectoderm, interactions of OV, SE and the surrounding tissues result in the formation of the lens placode within the head surface ectoderm. The concomitant invagination of lens placode and morphogenesis of the OV then lead to the formation of the lens vesicle in the center of the forming optic cup (Casey et al., 2021; Chow and Lang, 2001; Cvekl and Ashery-Padan, 2014; Gunhaga, 2011). Initially the lens vesicle is composed of lens progenitor cells. It is further patterned into lens epithelium at the anterior part of the lens, while posteriorly located progenitors differentiate into lens fiber cells (Kallifatidis et al., 2011; Zhou et al., 2006). The differentiation of lens fiber cells is characterized by the production and accumulation of crystallin proteins, the main structural proteins of the lens (Cvekl et al., 2015; Donaldson et al., 2017). This transition is marked by the extensive elongation of cells (Cheng et al., 2017; Fudge et al., 2011; Rao and Maddala, 2006) and the subsequent degradation of all organelles and de-nucleation (Bassnett, 2009; Brennan et al., 2021; Chaffee et al., 2014; Wride, 2011), minimizing light scattering and providing optimal refractive properties, all together resulting in the optical properties of the lens (Bassnett et al., 2011).

Several studies have shown that the conserved process of lens formation depends on the intricate interplay of intrinsic factors and extracellular signalling pathways (Cvekl and Ashery-Padan, 2014; Cvekl and Zhang, 2017; Lang, 2004; McAvoy et al., 1999; Ogino and Yasuda, 2000). The transcription factors *Pax6*, c-*Maf*, *Sox1*, *Prox1*, *Hsf4*, *Six3*, *Foxe3*, *Pitx3*, *Sox2* characterise early lens development and are directly involved in the processes of initial lens morphogenesis and lens fiber cell differentiation (Cvekl and Duncan, 2007; Cvekl and Zhang, 2017; Lang, 2004; Ogino et al., 2012; Ogino and Yasuda, 2000). Establishment of lens-competent progenitors as well as differentiation of lens fiber cells relies on the interplay of the FGF, BMP, and WNT signalling pathways (Faber et al., 2001, 2001; Furuta and Hogan, 1998;

Grocott et al., 2011; Gunhaga, 2011; Jarrin et al., 2012; Kreslova et al., 2007; Le and Musil, 2001; Litsiou et al., 2005; Lovicu and McAvoy, 2005; McAvoy et al., 1999; Rajagopal et al., 2009).

Although many aspects of lens formation are well characterized, addressing the molecular mechanisms in the complex environment of the developing embryo is often challenging. Advances in stem cell and developmental biology allowed the introduction, fast development and prominent use of organoids as new models to study complex processes of *ex vivo* organ formation. Those allow to gain insight into cellular and molecular processes in normal and diseased conditions in a controlled and experimentally accessible *in vitro* environment (Chen et al., 2024; Hofer and Lutolf, 2021; Kretzschmar and Clevers, 2016; Sahu and Sharan, 2020; Zhao et al., 2022). Mouse, human and fish pluripotent embryonic stem cells have been shown to self-organise into retinal tissue when aggregated and cultured under minimal growth factor 3D suspension culture conditions. Retinal organoids recapitulate key aspects of embryonic development with respect to cell fate commitment, tissue patterning, cell differentiation and physiological functions (Eiraku et al., 2011; Kuwahara et al., 2015; Nakano et al., 2012; Völkner et al., 2016; Zilova et al., 2021). Since the culture conditions for the generation of retinal organoids favor neuroectodermal differentiation over non-neuronal ectodermal fates, they did so far not allow for concomitant lens formation (Eiraku et al., 2011; Kuwahara et al., 2015; Nakano et al., 2012; Zilova et al., 2021).

Generation of lens-like structures (lentoid bodies) by stepwise differentiation of human ES or iPS cells has been reported (Cvekl and Camerino, 2022; Dincer et al., 2013; Fu et al., 2017; Leung et al., 2013; Yang et al., 2010). However, most lens differentiation protocols directly favor the promotion of placodal fate on the expense of the neuronal fate (Dincer et al., 2013; Fu et al., 2017; Yang et al., 2010). This rendered the generation of more complex organoids composed of cell types of diverse embryonic origin challenging. Interestingly, two independent studies had reported the appearance of lens marker-expressing cells in close vicinity to retina tissue in 3D organoid culture (Gabriel et al., 2021; Mellough et al., 2015). However, considering the complex nature of tissue interactions required for the coordinated formation of both lens and retina *in vivo*, this raises the intriguing question of how a lens is formed in neuronal organoids and to what extend that process recapitulates *in vivo* organ formation.

Here we used organoids derived from *Oryzias latipes* blastula cells as a model system to investigate the formation of ocular organoids composed of both, retina and lens. By following the individual populations of lens– and retina-committed progenitors, we found that similar to the *in vivo* situation, lens and retina originate from spatially confined and separated populations of progenitor cells. The subsequent differentiation into their respective lineages was found to be dependent on BMP and FGF signaling pathways. Their differentiation and morphogenesis are characterized by the expression of key regulatory factors implicated in eye (lens and retina) organogenesis, however following an entirely different morphogenetic route towards forming an eye cup enclosing a centrally positioned lens.

Results

Fish retinal organoids form embedded lenses under specific culture conditions

We have previously reported the generation of retinal organoids from fish pluripotent embryonic cells grown in 3D suspension culture in minimal growth factor-containing media. Such organoids acquire anterior neural fate and consist of retina and presumptive forebrain tissue (Zilova et al., 2021). Here, we modified our previously described protocol (**Figure 1a**) in order to generate organoids, that differentiated into retinal and lens tissues.

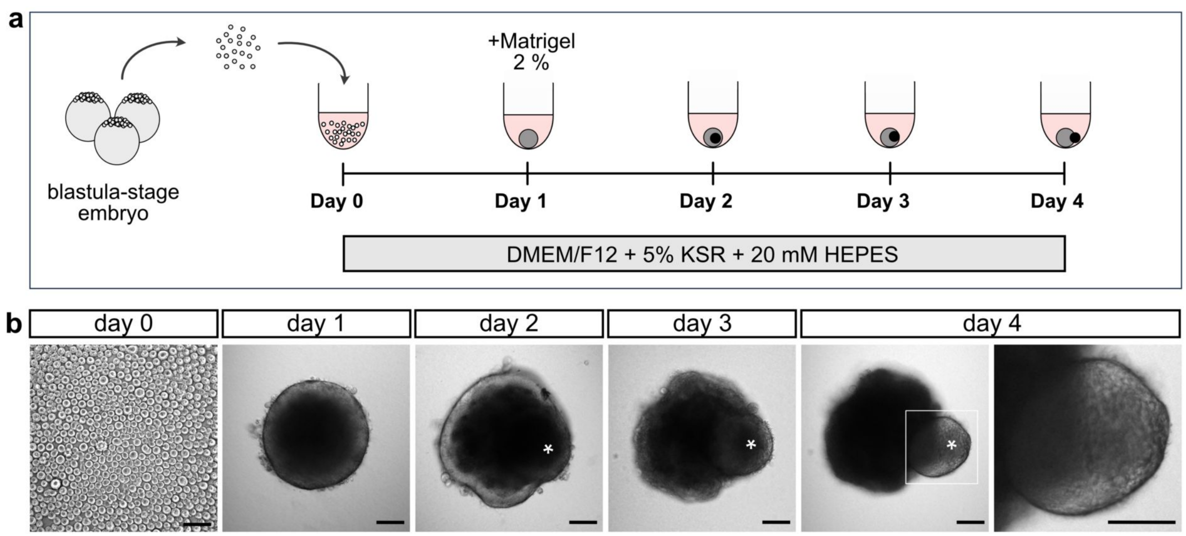

Figure 1.

Generation of organoids from medaka pluripotent embryonic cells.

(a) Schematic representation of organoid generation. At day 0, pluripotent embryonic cells were extracted from blastula-stage medaka embryos and seeded in U-bottom, low-binding 96-well plate into DMEM/F12 media supplemented with 5 % KSR and 20 mM HEPES pH 7.4. At day 1 aggregates were supplemented with 2 % Matrigel. **(b)** Bright-field pictures of indicated stages of organoid formation showing the appearance of lens-like structure (asterisk). KSR, knockout serum replacement. Pictures acquired with wide-field microscope (Mi8; Leica). Scale bar 100 μm.

Aggregation of fish primary pluripotent cells in DMEM/F12-based differentiation media supplemented with 5% knockout serum replacement and 20 mM HEPES pH 7.4 (see Materials and Methods for detailed composition) resulted in the efficient formation of retinal organoids containing a single transparent, spherical lens-like structure surrounded by neuronal tissue (**Figure 1b**). The lens-like structure was clearly visible at day 4 but first signs of its formation could be observed from day 2 onwards (**Figure 1b**, asterisk). We showed that organoids were composed of both retina and lens tissue, arranged in a way that was reminiscent of the tissue arrangement in the embryonic eye by using reporter lines and molecular markers. Differentiated lens cells expressing crystallin were labelled in the *Gaudi*^{RSG} transgenic reporter line (Centanin et al., 2014) that carries a *crystallin::ECFP* (*Cry::ECFP*) reporter. Lens fiber cells were identified with the "lens fiber cell" marker antibody ZL-1 (referred to as LFC), and early and late lens cells were identified with an anti-Prox1 antibody (**Figure 2a** and **2b**). Prox1 is expressed by cells of the lens placode and differentiating lens fiber cells of different species including mouse, zebrafish and medaka (Mikula Mrstakova and Kozmik, 2024; Pistocchi et al., 2008; Wigle et al., 1999). Anti-LFC labels lens fiber cells in zebrafish and is suggested to target one of the crystallin proteins (Greiling et al., 2010; Shi et al., 2006). In the medaka embryo, the expression of both, crystallin promoter-driven ECFP and LFC marker was observed exclusively in the lens 3 days post-fertilization (dpf) (**Figure 2a**). Prox1 immunolabeling was detected in the lens and a subpopulation of differentiating retinal cells in the central region of the retina (**Figure 2a**). The expression of all three lens markers was detected in day 4 ocular organoids (**Figure 2b** and **2c**), demonstrating that the spherical structure formed by the organoids is indeed the lens. ECFP labeling was sparse (25 % *Cry::ECFP*, **Figure 2b**) reflecting the mode of generation of the organoids by a combination of 25 % cells carrying the *Cry::ECFP* reporter and 75 % wild-type cells (see Materials and Methods for details).

The transgenic reporter lines *Atoh7::EGFP* (Del Bene et al., 2007) and *Rx3::H2B-GFP* (Rembold et al., 2006) were used to identify retinal tissue in lens carrying organoids. In embryos *Atoh7::EGFP* activity was confined to the retina, labelling differentiating retinal neurons at 3 dpf (**Figure 2a**). In organoids, labeled retinal cells surrounded the forming lens, reminiscent of the situation in the eye. To address the distribution of retinal and lens cells we used blastula cells from *Atoh7::EGFP* and *Rx3::H2B-GFP* retina-specific reporter lines (**Figure 2d** and **2e**) and co-labeled *Rx3::H2B-GFP*-derived day 4 organoids with anti-LFC antibody (**Figure 2e**).

Analysis of the organoid lens revealed a great similarity with structural features of the developing embryonic lens, such as the elongation of Prox1^{+}/LFC^{+} cells, typical for differentiating lens fiber cells and atypically looking nuclei indicating disorganization of nuclear material, a process tightly connected to lens fiber cell differentiation (**Figure 2b** and **2h**, arrowhead).

In 462 organoids derived from 38 independent experiments we observed lens formation, in 83.5 % (n=386) of organoids. A careful evaluation of different culture conditions and their impact on the lens formation revealed that lens formation depends on supplementation of the DMEM/F12-based differentiation media with 20 mM HEPES buffer (Figure S1a and 1b). While organoids grown without HEPES supplementation never formed lenses (n=63 organoids in 6 independent experiments, see also Zilova et al., 2021), 92 % of organoids grown in media supplemented with HEPES (n=83 organoids in 6 independent experiments) formed lenses (Figure S1b).

We tested the impact of extracellular matrix (ECM) supplementation on lens formation, as existing protocols for generation of lentoid bodies from human pluripotent embryonic stem cells mostly rely on supplementation with ECM proteins (Dincer et al., 2013; Fu et al., 2017; Leung et al., 2013; Yang et al., 2010), usually in form of Matrigel. Interestingly, lens cell fate specification as well as lens fiber cell differentiation, judged by the expression of Prox1 and LFC, occurred independently of Matrigel (2 % final concentration) supplementation in medaka-derived organoids (Figure S1c).

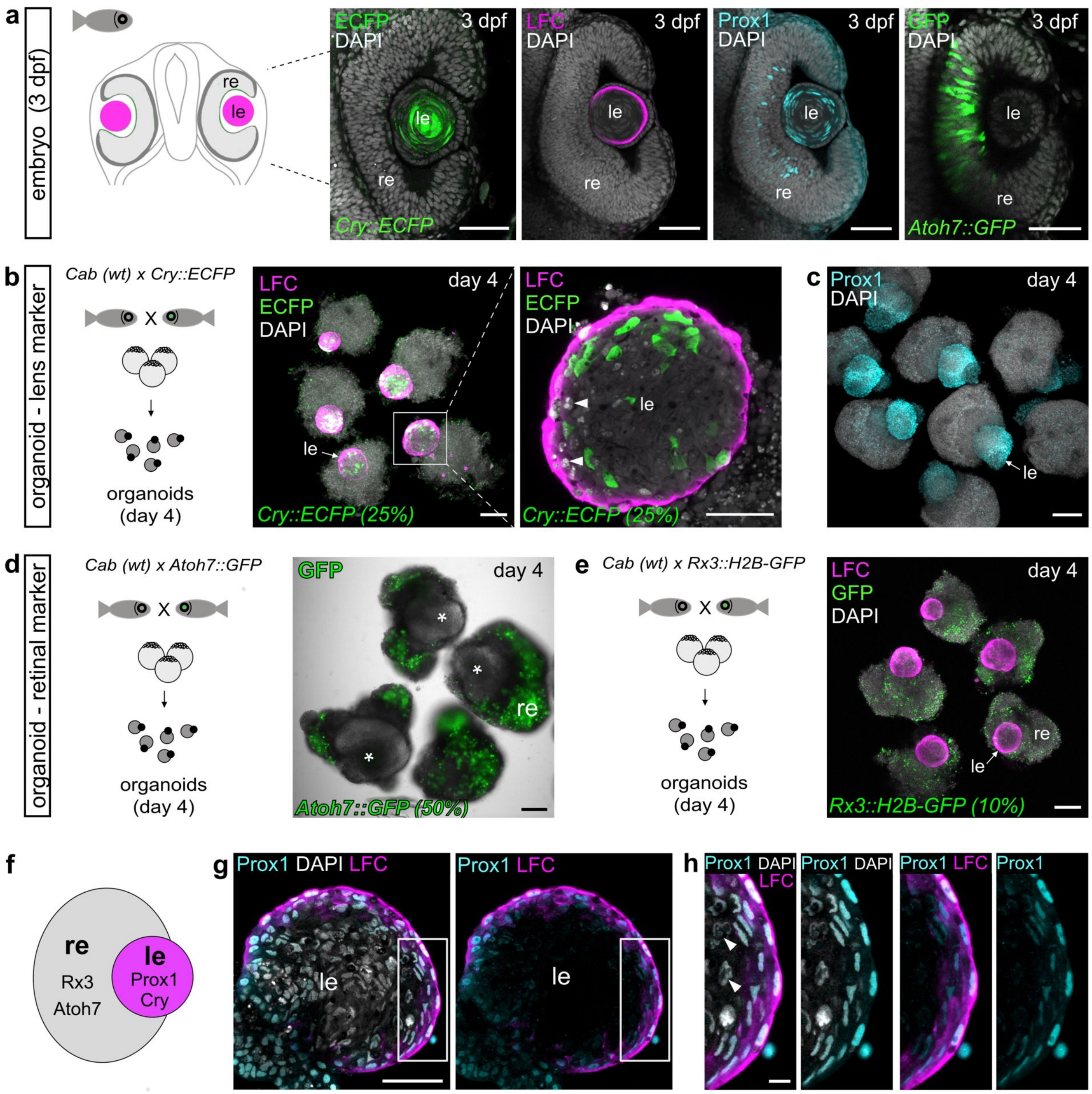

Figure 2.

Fish retinal organoids form lenses.

(a) Expression of lens-specific markers (*Cry::ECFP*, lens fiber cell (LFC) and Prox1) and retina-specific marker (*Atoh7::EGFP*) in medaka embryonic eyes at 3 dpf. Optical sections showing the expression of indicated markers detected by whole-mount immunohistochemistry with the use of anti-GFP (green), anti-LFC (magenta) and anti-Prox1(cyan) antibodies, co-labelled with DAPI nuclear stain (grey). **(b)** Organoid generation from crystallin reporter line (*Cry::ECFP*). Expression of lens-specific markers (*Cry::ECFP* and LFC) in day 4 organoids detected by anti-GFP and anti-LFC antibodies. Represented by maximal projection for overview and single optical section for the detail of the lens. **(c)** Expression of Prox1 in day 4 organoids derived from wild-type medaka cells detected with anti-Prox1 antibody, co-stained with DAPI nuclear stain; displayed as maximal z-projection. **(d)** Organoid generation from retina-specific reporter line (*Atoh7::EGFP*). Overlay of bright-field and wide-field images showing EGFP fluorescence in *Atoh7*-expressing cells, lens indicated with asterisk. **(e)** Organoids generated from retina-specific reporter line (*Rx3::H2B-GFP*) labeled with anti-GFP and anti-LFC antibody displayed as maximal z-projection. **(f)** Schematic representation of distribution of retina– and lens-specific markers in day 4 organoids. **(g)** Optical section of day 4 organoid lens labeled with anti-Prox1 and anti-LFC antibodies, co-labeled with DAPI, showing elongated Prox1^{+}/LFC^{+} cells on the surface of the lens. **(h)** Enlargement of indicated area in g. For *Cry::ECFP*, *Atoh7::EGFP* and *Rx3::H2B-GFP*. The percentage (%) indicates how many cells carry the indicated transgene. Arrow in b, c and e indicates position of the lens. Arrowheads in b and h indicate abnormally looking nuclei. le, lens; re, retina; wt, wild-type; LFC, lens fiber cell. Scale bar 50 μm in a, g and enlargement of b, 100 μm in b-e, 10 μm in h.

The lens originates from a spatially distinct population of placodal progenitors located in the central region of the organoid

We further investigated the origin of lens cells and how its formation compares to lens development *in vivo* in the developing organism. In the embryo, the lens originates from the lens competent head surface ectoderm (SE, magenta), which directly abuts the evaginating optic vesicle (OV, prospective retina, green) (**Figure 3a**). OV and SE invaginate in a coordinated way to form a lens sphere surrounded by the retina. In medaka embryos, those processes occur between 1 dpf and 2 dpf (**Figure 3a**, 3b). Concomitant with the completion of lens formation, the differentiation of lens fiber cells is initiated and marked by the expression of crystallin proteins (LFC; cyan) (**Figure 3b**). To track the distribution of lens and retinal progenitors, we generated organoids from the *Rx3::H2B-GFP* reporter line, which highlights retina-committed cells by nuclear GFP expression (Figure S2a and 2b). To visualize lens-committed progenitor cells within the SE (Mikula Mrstakova and Kozmik, 2024), we co-labeled those organoids with an anti-Prox1 antibody (Figure S2b) and followed the expression of *Rx3*-specific GFP and Prox1 from day 1 to day 2 (**Figure 3c**). Retinal progenitor cells were distinctly localized in the cortical region around the surface of day 1 organoids (Figure S2d, **Figure 3c**, Video S1), while lens progenitors (expressing Prox1) were not detectable at this stage (**Figure 3c**). Expression of retinal markers was confined exclusively to the cortex and was never detected in the central part of the organoid (**Figure 2b**, Figure S2d, Video S1). From day 1.5 onwards (stage S22 – 1 day and 14 h) (Iwamatsu, 2004), embryos as well as organoids (30 hours post aggregation) started expressing Prox1 (**Figure 3b** and **3c**). In the embryo, Prox1 expression was localized to the surface of the head, labeling the presumptive lens region (**Figure 3b**). In the organoid, the expression was confined to the central core of the organoid with Prox1^{+} cells organized into a sphere (**Figure 3c**). Prox1 expression in the central core of the organoids was followed by the expression of lens fiber cell differentiation marker LFC at day 2, concomitant with the onset of Prox1 and crystallin expression in the embryo (**Figure 3b** and **3c**).

To specifically address the dynamics, onset, and cellular re-arrangements of lens-committed progenitors, we generated a *Foxe3::GFP* reporter line. *Foxe3* (winged helix/forkhead domain transcription factor) is necessary for proper lens development and is expressed by the cells of the presumptive lens ectoderm and lens epithelium of different species, including medaka (Blixt et al., 2000; Brownell et al., 2000; Mikula Mrstakova and Kozmik, 2024; Shi et al., 2006). This pattern is fully recapitulated by *Foxe3::GFP* with an onset of expression starting at 1 dpf (stage S20/S21; Iwamatsu, 2004). It was further expressed by the majority of lens cells at 2 dpf, where it preceded the expression of lens fiber cell marker LFC (**Figure 3d**). During development, the lens is composed of anteriorly located lens epithelium (proliferative lens progenitors) and posteriorly and centrally located differentiating lens fiber cells (Greiling et al., 2010). We could differentiate those parts of the developing lens with *Foxe3::GFP* expression in both, lens fiber cells and lens epithelium in 3 dpf embryo, and the expression of Prox1 and LFC restricted to differentiating fiber cells, but excluded from the lens epithelium (Figure S3a, arrows). Analysis at later stages showed that at 4 dpf, *Foxe3::GFP* expression was maintained in the lens epithelium (proliferatively active progenitors) and was nearly undetectable in differentiated lens fiber cells (Figure S3b, arrows).

In organoids, lens progenitor cells expressing *Foxe3::GFP* were first found in the central core at day 1 (26 hours post aggregation) (Video S2, **Figure 3e**, **Figure 4**, Video S3). These cells appeared in a pattern complementary to the retinal marker *Rx3::H2B-GFP* (Video S1), indicating that lens progenitor cells originate from a spatially distinct population in the central core of the organoid. We never detected any *Rx3*-positive cells eventually acquiring a lens fate. To address the distribution of lens and retina domains, we generated organoids from mixed blastula cells derived from the retina-specific reporter line *Rx2::H2B-RFP* (Inoue and Wittbrodt, 2011) and the lens specific reporter line *Foxe3::GFP* (Figure S3c). By day 2, lens progenitor cells expressing *Foxe3::GFP* contributed to the formation of a spherical lens in the central core of the organoid surrounded by

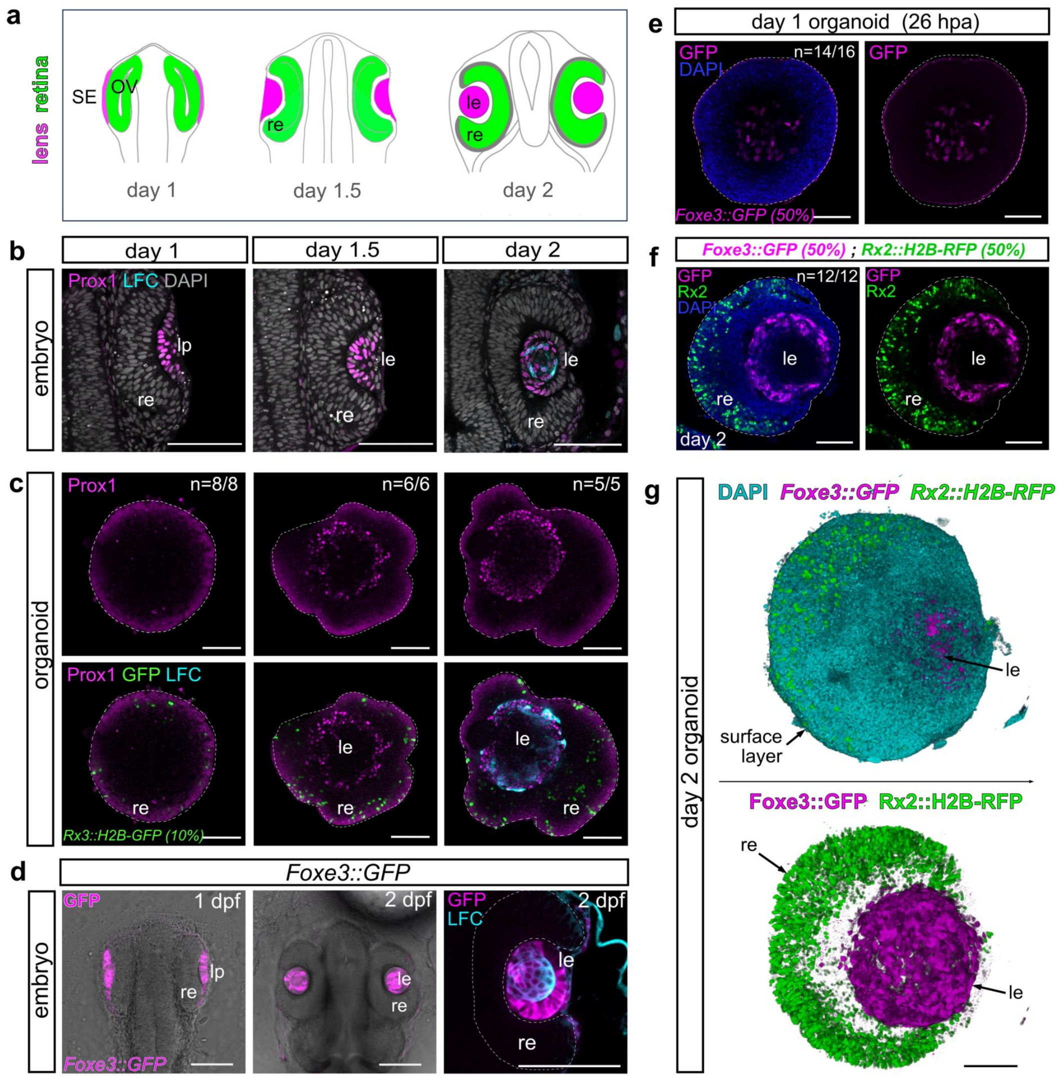

Figure 3.

The lens originates from a spatially separated population of lens progenitors in center of the organoid.

(See also Figure S2 and S3, Video S1, S3 and S4.) **(a)** Schematic representation of lens formation in medaka. The lens develops from lens-competent head surface ectoderm (SE, magenta), retinal cells from the OV (green). Surface ectoderm forms the lens placode (lp) that further invaginates together with the OV to form the optic cup with a lens sphere surrounded by the retina. **(b)** Optical sections through wild-type medaka embryos at indicated stages stained with the antibody against Prox1, a lens placode-specific marker and LFC, a lens fiber cell marker. **(c)** Optical sections through organoids derived from the *Rx3::H2B-GFP* reporter line stained with antibody against Prox1 (magenta) and LFC (cyan). **(d)** Lens specific GFP expression in embryos of the *Foxe3::GFP* reporter line (magenta). Optical section showing the overlay of bright-field and GFP fluorescence in the 1 dpf and 2 dpf medaka embryos (left and middle). Detail of the eye (right) showing the expression domain of *Foxe3::GFP* (magenta) and LFC (cyan) detected by immunohistochemistry. **(e)** Optical section through a day 1 organoid (26 hpa) derived from *Foxe3::GFP* reporter line showing the lens-specific expression of GFP (magenta) detected by anti-GFP antibody, co-stained with DAPI (blue). **(f)** Optical section through a day 2 organoid derived from 50 % *Foxe3::GFP* and 50 % *Rx2::H2B-RFP* reporter blastula cells. Expression of RFP (green) and GFP (magenta), detected by antibodies is mutually exclusive, co-stained with DAPI (blue). **(g)** 3D rendering of *Rx2::H2B-RFP* (green), *Foxe3::GFP* (magenta) organoid shown in f, DAPI stained nuclei are in cyan. n numbers in c and e indicate the number of analyzed organoids with described phenotype. LFC, lens fiber cell; le, lens; re, retina; lp, lens placode; SE, surface ectoderm; OV, optic vesicle; dpf, day post fertilization; hpa, hours post aggregation. Dashed line in c, e and f indicates the outline of the organoid. Dashed line in d indicates the outline of the retina. Scale bar 100 µm.

unlabeled cells and eventually by the cortical retinal tissue (**Figure 3g**, Video S4). This was confirmed by time-lapse imaging of ocular organoid development from day 1 to day 2 where *Foxe3::GFP*-expressing cells appear initially in the core region of the organoid where they organize into a spherical lens within 16 hours (Video S2, **Figure 4**). This process was accompanied by gradual displacement of the forming lens from center to the periphery where the lens is eventually reaching the surface of the organoid (Video S2, **Figure 3f** and **3g**, **Figure 4**).

Further analysis showed that the organoid lens consists of proliferating as well as differentiating cells (Figure S3d and 3e), however not in distinct domains as in the embryo. We found that *Foxe3*-driven GFP was co-expressed with the LFC differentiation marker in day 2 organoids (Figure S3e). EdU incorporation also showed that the lens is formed by proliferative (EdU^+) *Foxe3::GFP*-expressing cells (Figure S3d). On the surface of day 2 organoid lenses, we initially detected stretches of $Foxe3^+/LFC^-$ cells, reminiscent of the lens progenitor cells of the lens epithelium (Figure S3e and 3f, arrows). However, at day 4, organoid lenses did not show any apparent lens epithelium ($Prox1^-/LFC^-$) (**Figure 2g** and **2h**). In fact, we did not detect any $Prox1^-/LFC^-$ cells on the surface of the day 4 lenses, indicating that in organoid lenses a lens epithelium is absent at later stages.

Lens size scales with organoid size

Since retinal and lens progenitors occupied distinct regions within the organoid, we next investigated how their distribution is affected by total organoid size. We generated organoids from different numbers of cells at day 0 (4,000, 2,000, 1,000 and 500 cells). This approach resulted in organoids of different sizes at day 1: organoids derived from 4,000 cells had an average diameter of 521 ± 13 µm, those from 2,000 cells measured 379 ± 15 µm, from 1,000 cells 310 ± 22 µm, and from 500 cells 201 ± 55 µm (**Figure 5a**, 5b and 5d). Interestingly, the thickness of the tissue acquiring retinal progenitor cell identity in the cortex of the ocular organoid (*Rx3::H2B-GFP*) remained constant, independent of the initial size of the aggregate reaching to the depth of: 59.7 µm ± 1.6 µm for 4,000 cells; 55.6 ± 3.1 µm for 2,000 cells; 56.4 µm ± 3.2 µm for 1,000 cells and 55.4 µm ± 6.8 µm for 500 cells (**Figure 5b** and **5e**). While the thickness of retinal neuroepithelium stayed constant, the lens size scaled proportionally with the size of the organoid. By day 4, larger organoids developed larger lenses, whereas smaller organoids formed smaller lenses (224 µm ± 12 µm for 4,000 cells; 199 µm ± 26 µm for 2,000 cells; 115 µm ± 33 µm for 1,000 cells and 68 µm ± 20 µm for 500 cells) (**Figure 5c** and **5f**). Although 500 cell derived organoids were able to form lenses, the efficiency of lens formation was considerably lower (33.3 %; **Figure 5g**) than in organoids of the bigger sizes (between 86.7 – 95 % for organoids generated by 1,000 or more cells) (**Figure 5g**). Ocular organoids generated by the aggregation of less than 500 cells did not form any lenses (**Figure 5g**). Taken together, these data indicate that lens size scales with the size of the organoid and is defined by the constant thickness of the forming retina.

Lens formation depends on temporal activity of FGF and BMP signaling

The competence to form a lens is established during the formation of the anterior neural plate within the region of the anterior neural plate border and is regulated by the coordinated activity of BMP and FGF signaling (Gunhaga, 2011; Litsiou et al., 2005; Sjödal et al., 2007). Once lens competence is established within the SE, FGF and BMP signaling induce its differentiation into the lens placode and subsequently into the lens, accompanied by the expression of structural lens proteins (Faber et al., 2001; Furuta and Hogan, 1998; Le and Musil, 2001; Rajagopal et al., 2009; Sjödal et al., 2007; Wawersik et al., 1999; Zhao et al., 2008). To address the contribution of BMP and FGF signaling in organoid lens formation, we exposed developing organoids to FGF and BMP antagonists, SU5402 and Noggin or Dorsomorphin, respectively. To separate the establishment of lens competence from the process of lens differentiation, we divided the treatments into two discrete time windows: day 0 to day 1 corresponding to the time before the

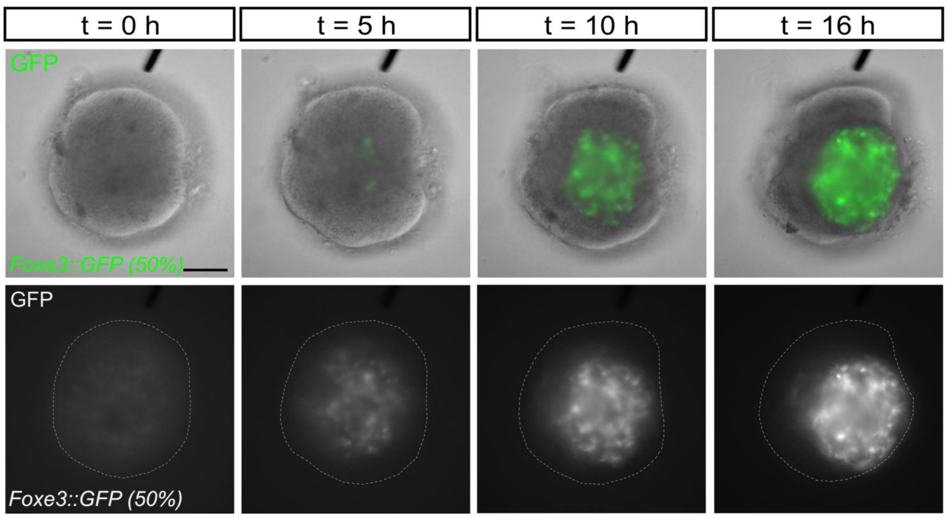

Figure 4

Dynamics of lens formation in ocular organoids.

Single organoid (from Video S2) generated from *Foxe3::GFP* reporter (50 %) and wild-type blastula cells imaged from day 1 (24 h post aggregation; t=0 h) to day 2 (40 h post aggregation, t=16 h). Overlay (top panel) of bright-field and GFP fluorescence and GFP fluorescence (bottom panel, gray) of indicated timepoints between day 1 and day 2 organoid development. Dashed lines indicate the outline of the organoid. Scale bar 100 µm.

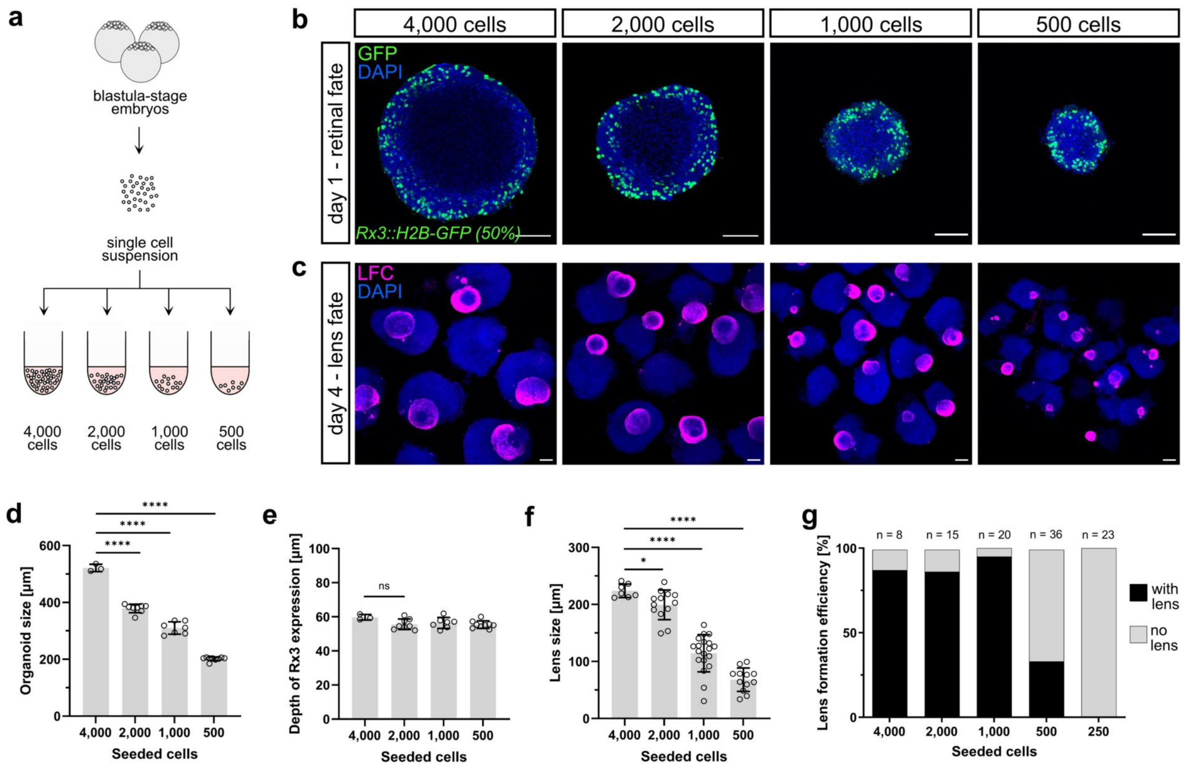

Figure 5.

Lens size scales with organoid size.

(a) Schematic representation of generation of organoids of different sizes. Pluripotent embryonic cells were seeded in different seeding densities. **(b)** Representative optical sections of day 1 organoids generated by aggregation of indicated number of cells from *Rx3::H2B-GFP* reporter line immunolabeled with anti-GFP antibody, co-stained with DAPI. For *Rx3::H2B-GFP*, the percentage (%) indicates how many cells carry the indicated transgene. **(c)** Representative images of day 4 organoids generated by indicated number of cells stained with an anti-LFC antibody (maximal z-projections). **(d)** Quantification of diameter of day 1 organoids generated by aggregation of indicated number of cells. **(e)** Quantification of depth of retina-specific marker expression (*Rx3::H2B-GFP*) at day 1 generated by aggregation of indicated number of cells. Number of analyzed organoids: n=3 for 4,000 cells, n=8 for 2,000 cells, n=7 for 1,000 cells, n=9 for 500 cells. **(f)** Quantification of lens size measured as diameter of the lens in day 4 organoids. n=7 for 4,000 cells, n=13 for 2,000 cells, n=19 for 1,000 cells, n=19 for 500 cells. **(g)** Efficiency of lens formation in organoids of different sizes measured as proportion of organoids carrying lens judged by immunolabeling with anti-LFC antibody. Statistical significance was calculated using an unpaired two-tailed t-test. n.s., not significant; ****p<0.0001, *p<0.05. le, lens; re, retina; LFC, lens fiber cell. Scale bar 100 µm.

lens placode formation (establishment of lens competence), and day 1 to day 2 covering lens placode formation and differentiation of lens fiber cells (**Figure 6a**). We then assessed the organoids at day 2 for the expression of the early lens progenitor marker *Foxe3::GFP* and the differentiation marker LFC.

Blocking BMP signaling with Noggin or Dorsomorphin completely prevented the expression of lens-specific markers if applied from day 0 to day 1 (**Figure 6b** and **6c**) but did not impact on lens formation if administered from day 1 to day 2 (**Figure 6b** and **6d**). In contrast, inhibition of FGF signaling with SU5402-mediated did not affect initial lens formation if applied from day 0 to day 1 and allowed expression of both, lens placode and lens differentiation markers. However, it efficiently blocked the differentiation of lens fiber cells when applied from day 1 to day 2, and completely prevented the expression of the LFC marker (**Figure 6b**, 6c and 6d). This indicates that similar to embryonic development, the establishment of lens competence (between day 0 and day 1) depends on the activity of BMP signaling, while FGF signaling is required for the lens differentiation process (day 1 to day 2).

Lens organoid gene expression recapitulates the onset of lens-specific genes *in vivo*

To address the temporal dynamics of lens-specific gene expression in *in vivo* and in organoid-based lens development, we performed total RNA sequencing analysis. Embryos at 1, 2, 3, and 4 dpf as well as organoids at corresponding developmental stages, were analyzed for the expression of key markers of lens development. For total RNA sequencing we extracted RNA from whole organoids (composed of both lens and retina), the anterior part of day 1 and 2 embryos (dissected right behind optic vesicles or optic cups), and the eyes of 3 dpf and 4 dpf embryos (**Figure 7a**). Within the total RNA reads we focused specifically on key genes implicated in the different stages of lens development. Among the genes analysed, the earliest expressed was the transcription factor *Foxe3*, with expression peaking at day 1 of development, followed by the upregulation of the transcription factor *Pitx3* (**Figure 7c**, Data S1). Day 2 marks the onset of expression of main structural lens proteins in embryos and organoids, namely: crystallins: *cryaa* (αA-crystallin), *cryba1a* (βA1-crystallin), *crybb1l2* (βB1 like 2), *crybb1l3* (βB1 like 3), *cryba2b* (βA2), *cryba4* (βA4), *crybb1* (βB1), *crygn2* (γN2-crystallins), *crygmx* (γMX), *crygm3* (γM3); major lens membrane structural proteins (MIPs/acquaporins) *mipa* and *mipb* coding for water channels ensuring water transport between lens cells (Agre and Kozono, 2003; Donaldson et al., 2024; Varadaraj et al., 1999); Gap junction proteins *Gja8* (connexin-50) and *Gje1* (connexin-23) and lens intrinsic membrane proteins *lim2.2* and *lim2.4* facilitating intercellular transport of metabolites and ions in lens fiber cells (Mathias et al., 2010). Day 3 was characterized by the onset of expression *hsf4* (heat shock transcription factor 4), coding for a transcription factor required for de-nucleation and organelle degradation during lens terminal differentiation (Cui et al., 2013; Gao et al., 2017). Accordingly, *dnase1l1l* (deoxyribonuclease I-like 1-like) and *plaat1* (phospholipase A/acyltransferase) implicated in DNA and lens organelle degradation, respectively (Morishita et al., 2021; Zhang et al., 2020) were also expressed from day 3 of development. At the same stage, the expression of proteins making up the structural scaffold critical for fiber cell architecture and mechanical properties of the lens was initiated. Those include beaded filament proteins *Bfsp1* (filensin) and *Bfsp2* (phakinin) and *Lgsn* (lengsin) (Harding et al., 2008; Perng et al., 2007; Song et al., 2009).

Our comparative analysis indicates highly reminiscent gene expression patterns of key marker genes when comparing organoid and embryonic lens development (**Figure 7c**, Data S1). This demonstrates that our conditions promote the development of ocular organoids in which the expression of key markers genes is temporally recapitulating the order of expression in the development of the embryonic lens.

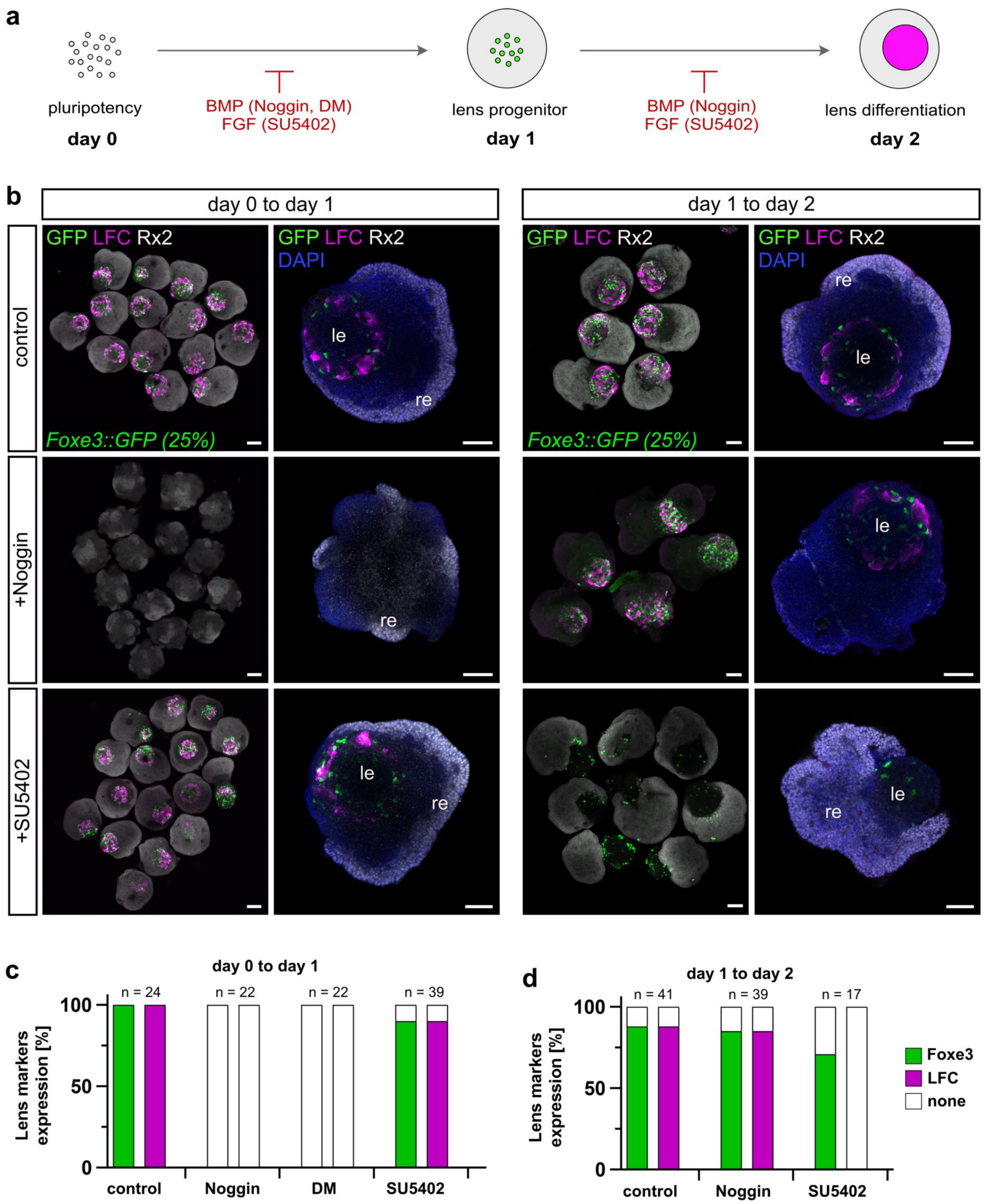

Figure 6.

Lens formation is dependent on temporal activity of FGF and BMP signaling.

(a) Schematic representation of organoid treatment with FGF and BMP signaling antagonists. Organoids were generated from *Foxe3::GFP* reporter line. Inhibitors of BMP (Noggin, 100 ng/µl or Dorsomorphin –DM, 100 ng/ml) or FGF (SU5402, 10 µM) signaling were administered in time window from day 0 to day 1 or from day 1 to day 2. Organoids were analyzed at day 2 for the expression of lens progenitor marker *Foxe3::GFP* and lens fiber cell differentiation marker LFC. **(b)** Representative images of day 2 organoids treated as indicated and labeled with antibodies against GFP (green), LFC (magenta) and Rx2 (grey). Overview images are represented by maximal z-projections; single organoids are represented by single optical sections. For *Foxe3::GFP*, the percentage (%) indicates how many cells carry the indicated transgene. **(c)** Quantification of expression of *Foxe3::GFP* and LFC in control and treated organoids measured as proportion of organoids expressing lens progenitor marker *Foxe3::GFP* (green) and lens differentiation marker LCF (magenta). re, retina; le, lens; DM, Dorsomorphin; LFC, lens fiber cell. Scale bar 100 µm.

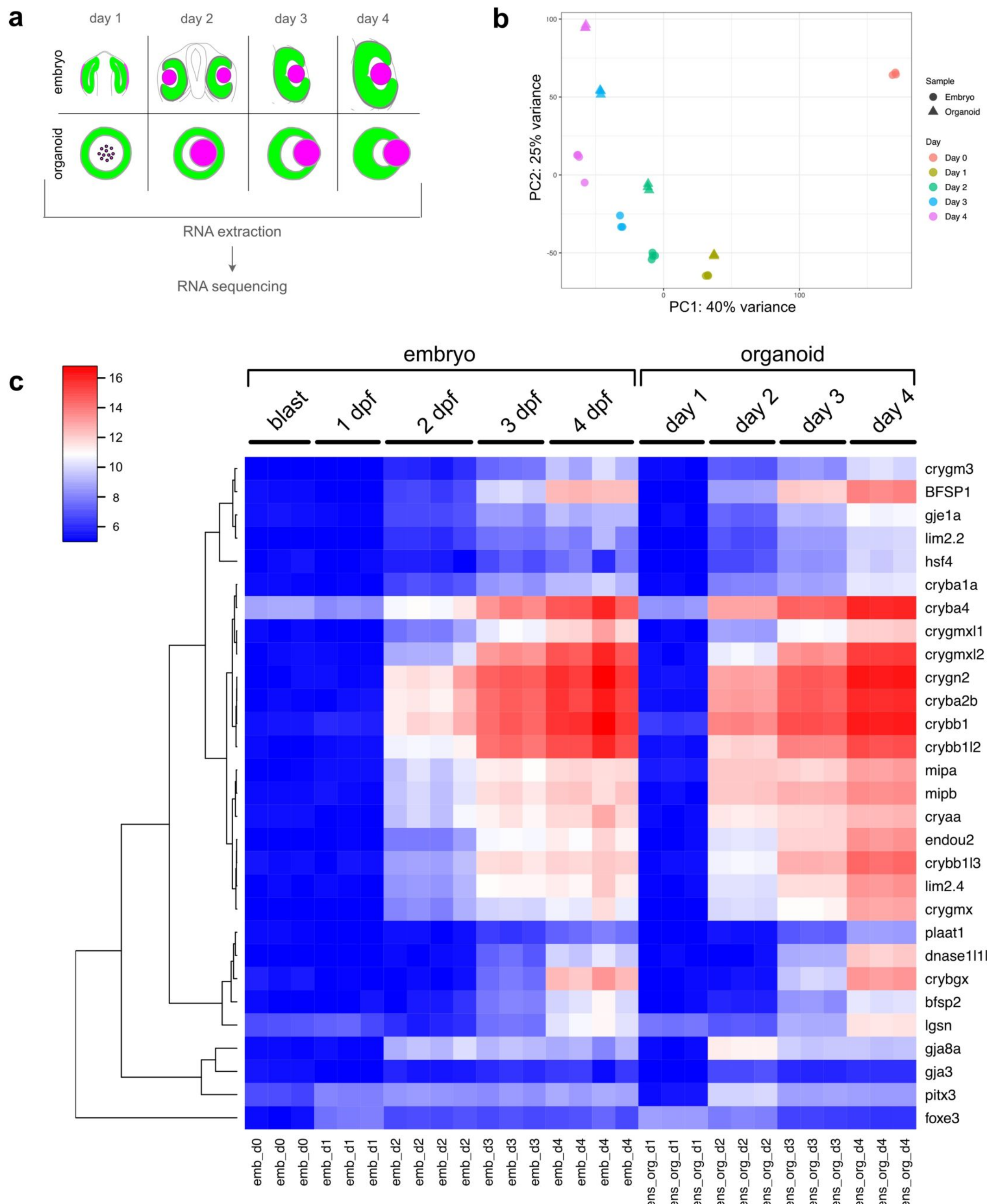

Figure 7.

Lens organoid gene expression recapitulates gene expression *in vivo*.

(a) Schematic representation of sample preparation. Whole organoids (composed of both lens and retinal tissue) were analyzed. For 3 dpf and 4 dpf medaka embryos, eyes were dissected (composed of both lens and retina). For 1 dpf and 2 dpf embryos the anterior part of the head right behind optic vesicles or optic cup, respectively was dissected and used for RNA isolation. **(b)** Principal component analysis of the transcriptomes of embryonic and organoid samples at the indicated stages. **(c)** Heatmap clustering of variance stabilized counts of lens-specific genes. blast, blastula-stage embryo; emb, embryo; lens_org, lens organoid.

Discussion

In vitro organoid development builds on the remarkable ability of embryonic stem cells or progenitor cells to autonomously self-organize in higher order structures with sophisticated microarchitecture, often closely resembling cell type compositions, 3D structure, and the function of the organ (Camp et al., 2015; Dye et al., 2015; Eiraku et al., 2011; Lancaster et al., 2013; McCracken et al., 2014; Nakano et al., 2012).

Here, we used medaka derived pluripotent cells as a model for the generation of complex ocular organoids composed of both, retina and lens. We showed that fish retinal organoids efficiently form embedded lenses, in a process that employs the molecular pathways including expression of key transcription factors and utilization of extracellular signaling pathways, as the embryo. In contrast to *in vivo* development, organoid cells did not follow the canonical process of lens formation, but used an alternative mode of cellular assembly to generate and grow the lens. While in the embryo, the lens sphere is shaped by the coordinated invagination of lens committed head surface ectoderm and retina-forming optic vesicle neuroepithelium, lens progenitors in the organoid were established in a different fashion directly inside of the retinal organoid to form the lens.

By following early progenitors of both retina (*Rx3*) and lens (*Foxe3*) we found that retina progenitor identity was acquired at the surface, while a spherical domain with lens progenitor identity was established in the center of the organoid (**Figure 3c** and **Figure 3e**). While in the embryo the lens originates from the head surface ectoderm, non-canonical routes to lens regeneration have been described in some species. In newts, the lens can be regenerated from the pigmented epithelial cells of the iris after injury or lens removal (Agata et al., 1993; Grogg et al., 2005; Maki et al., 2007). Lens formation was also observed in the cultures of chick neuroretinal cells (Iida et al., 2017; Okada et al., 1975) showing that retinal-fated cells can give rise to lens cells under certain circumstances. However, in medaka ocular organoids, retina and lens are formed by spatially separated populations of retina– and lens-committed progenitor cells.

The size of the lens scaled with the size of the organoid and was limited by the volume of the organoid that acquired retinal fate (**Figure 5**). The constant depth of retinal fated cells in the cortex under the surface of the organoid, regardless of the size of the aggregate, suggests that retinal fate is influenced by environmental gradients, which cause the diffusion-limited availability of retina-inducing factors. It has previously been shown that cells on the surface of the organoid experience higher level of oxygen tension, higher level of nutrients and signaling molecules than cells located more centrally (Grün et al., 2023; McMurtrey, 2016; Qian et al., 2019; Tse et al., 2021) when grown in 3D suspension culture. Additionally, cells at the surface experience different mechanical forces, mostly due to the interactions with ECM (Vining and Mooney, 2017). Both, retinal as well as lens fates, were established either before ECM addition or in the absence of Matrigel (Zilova et al., 2021, this study), making the environmental gradient more likely to contribute to the observed tissue arrangement in fish organoids.

Interestingly, Bmp inhibition in the early stages of organoid development, before the appearance of *Foxe3*-expressing lens progenitor cells (day 0 to day 1) completely abolished lens formation, while BMP inhibition during the lens formation process (from day 1 to day 2) did not show any impact. Thus, we hypothesize that a source of BMP in the forming neuroretina establishes initial lens competence in the center of the organoid. During normal development, BMP4 is secreted by the neuroepithelium of the developing OV and was found to be essential for lens induction (Furuta and Hogan, 1998; Rajagopal et al., 2009). Accordingly, ablation of the prospective neuroectoderm in chick (Kamachi et al., 1998) or defective OV formation in mouse (Klimova and Kozmik, 2014; Mathers et al., 1997; Porter et al., 1997) prevents lens formation, indicating that the developing neuroepithelium stimulates lens formation within the pre-placodal ectoderm. The emergence of lens fate in the center of ocular organoids consequently depends on a critical threshold of the morphogen to be reached in the center. The scaling of lens size with organoid size supports this hypothesis as larger organoids are better suited to reach this threshold – potentially due to increased total ligand production (in the forming neuroretina) resulting in the formation of a more pronounced gradient.

During normal development, developing retinal cells have been suggested to serve as a source of FGFs stimulating expression of lens structural genes and inducing lens differentiation (Chow et al., 1995; Faber et al., 2001; Gunhaga, 2011; Le and Musil, 2001; McAvoy et al., 1999; Robinson, 2006; Zhao et al., 2008). Accordingly, blocking of FGF signaling activity during organoid lens formation prevented lens differentiation. The dependency of lens formation on the activity of BMP and FGF signaling pathways has been well documented also in other *in vitro* systems where transient activation of BMP and FGF signaling is necessary for the *in vitro* differentiation of lens-like structures (lentoid bodies) (Dincer et al., 2013; Fu et al., 2017; Leung et al., 2013; Yang et al., 2010).

Since organoids show remarkable similarities to their *in vivo* counterparts when it comes to cellular composition and key structural features, we naturally tend to infer that corresponding structures are formed by corresponding cellular processes in both systems. Here we show an example of an alternative mode of cellular assembly that leads to a similar structural and molecular outcome.

Organoids can recapitulate many aspects of embryonic development. However, the absence of embryonic constraints can lead to significant differences in tissue organization, cellular interactions, and overall structure. While the goal of many organoid studies is to replicate *in vivo* conditions as closely as possible (Chen et al., 2024; Hofer and Lutolf, 2021; Kretzschmar and Clevers, 2016; Sahu and Sharan, 2020; Zhao et al., 2022), the unconstrained environment of organoids offers a unique opportunity to tap the organoid's potential of self-organization. This can uncover alternative developmental pathways that are not apparent *in vivo* and could expand our understanding of the plasticity, robustness and adaptability of developmental programs when unleashed from evolutionary constraints. Insights gained from alternative assembly processes could inform the design of synthetic biological systems and potentially lead to innovative strategies for creating complex tissues and organs *in vitro*.

Materials and Methods

Fish handling and maintenance

Medaka (*O. latipes*) stocks were maintained according to the local animal welfare standards (Tierschutzgesetz §11, Abs. 1, Nr. 1, husbandry permit AZ35-9185.64/BH, line generation permit number 35–9185.81/G-145/15 Wittbrodt). The following medaka lines were used in this study: Cab strain as a wild type (Loosli et al., 2000), *Rx3::H2B-GFP* (Rembold et al., 2006), Rx2::H2B-GFP (Inoue and Wittbrodt, 2011), *Atoh7::EGFP* (Del Bene et al., 2007), *Gaudi*^{RSG}(Centanin et al., 2014), *Foxe3::GFP* (this study).

Generation of organoids

Organoids were generated as previously described (Zilova et al., 2021) with slight modification of the protocol. Blastula stage embryos, stage 11 (Iwamatsu, 2004) were de-chorioneated, cells mass was isolated and washed twice with PBS (Thermo Fisher Scientific, Cat#:10010023). Cell mass was dissociated in PBS by pipetting up and down and centrifuged through 40 µm strainer (pluriSelect, Cat#:43-10040-51) for 3 min at 180 x g. Pellet was re-suspended in differentiation media: DMEM/F12 (Gibco, Cat#:11320033), 5 % KSR (Gibco, Cat#:10828028), 1x NEAA (Sigma-

Aldrich, Cat#:M7145), 1x sodium pyruvate (Sigma-Aldrich, Cat#:S8636), 1x penicillin/streptomycin, 20 mM Hepes pH 7.4 (Roth, Cat#:9105.4), 0.1 µM Φ3-mercaptoethanol (Gibco, Cat#:21985-023). If not stated otherwise, 3.000 cells per aggregate in 100 µl were seeded into individual wells of low-binding 96-well plate (Thermo, Cat#:174925 Nunclon Sphera). Cells were incubated in 26°C in the incubator without CO_2 control. At day 1, aggregates were washed in differentiation media, Matrigel (Corning, Cat#:356238) was added to final concentration of 2 % and aggregates were incubated in humidified tissue culture incubator under 5 % CO_2 condition.

To label specific population of cells, organoids were generated from fish reporter lines expressing fluorescent protein under the control of tissue specific promotor/enhancer (*Cry::ECFP*; *Rx3::H2B-GFP*; *Rx2::H2B-RFP*; *Atoh7::EGFP*; *FoxE3::GFP*). In that case, cells were extracted from the blastula stage embryos originating from the outcross of fish reporter line and wild-type fish, and in some cases further mixed with cells originating from wild-type embryos of the same stage to achieve sparse labeling. The proportion of cells coming for transgenic reporter line genetically able to express the transgene is indicated in the figures in percentage (%).

Treatment of cells/aggregates/organoids

Cells or aggregates were treated with Noggin (Merck Millipore, Cat#:GF173, 100 ng/ml), Dorsomorphin (Merck, Cat#:171260, 100 ng/ml), or SU5402 (Sigma, Cat#:SML0443, 10 µM). For treatment from day 0 to day 1, blastula stage cells were resuspended directly in differentiation media containing an antagonist in desired concentration. At day 1 (16 hours post aggregation), aggregates were washed twice with fresh differentiation media, supplemented with 2% Matrigel (Corning, Cat#:356238) in differentiation media and incubated till day 2. For treatment from day 1 to day 2, day 1 aggregates (16 hours post aggregation) were washed with differentiation media and transferred into antagonist-containing media and immediately supplemented with 2% Matrigel diluted in antagonist-containing media and incubated till day 2. All organoids were fixed at day 2 and analyzed for the presence of lens and retinal tissues by immunohistochemistry.

Fluorescent labeling

Fish embryos and organoids were fixed in 4% PFA overnight at 4°C, washed with PTW (PBS supplemented with 0.1 % Tween 20), permeabilized by incubation in acetone for 15 min in – 20°C and washed tree times with PTW. Samples were blocked in PTW supplemented with 4 % sheep sera (Sigma, Cat#:14C509), 1 % BSA (Sigma, Cat#:4503) and 1 % DMSO (Sigma, Cat#:D4540) and incubated with primary antibody diluted in 0.1 x blocking solution overnight at 4°C. Primary antibodies used in this study: rabbit anti-Rx2 (Reinhardt et al., 2015) (1:500), chicken anti-GFP (Invitrogen, Cat#:A10262, 1:500), mouse anti-lens fiber cell (Abcam, Cat#:ab185979, ZL-1, 1:500), rabbit anti-Prox1 (Millipore, Cat#:AB5475, 1:2000), rabbit anti-RFP (Clontech, Cat#:632496, 1:500). Samples were washed with PTW and incubated with the mixture of DAPI nuclear stain (Sigma, Cat#:D9564, 1:500) and secondary antibodies diluted in 0.1 x blocking solution overnight at 4°C. Secondary antibodies used in this study: donkey anti-mouse 647 (Jackson, Cat#:105869, 1:750), donkey anti-chicken 488 (Jackson, Cat#:703-545-155, 1: 750), donkey anti-rabbit 594 (Invitrogen, Cat#:A21207, 1:750). Samples were washed with PTW. For detection of proliferating cells, Click-iT EdU Alexa Fluor 647 Imaging Kit (Invitrogen, Cat#:C10340) was used according to manufacturer instructions. All immunolabeled samples were transferred into clearing solution composed of: 20% urea (Vetec, Cat#:V900119), 30% D-sorbitol (Vetec, Cat#:V900390), 5% glycerol (Vetec, Cat#:V900122) in DMSO (Vetec, Cat#:V900090) (Zhu et al., 2019) and imaged with laser scanning confocal microscope Sp8 (Leica).

Imaging

Fixed immunolabeled samples were imaged in MatTek glass bottom dish (Mattek, Cat#:D35G-1.5-10-C) with Sp8 confocal microscope (Leica). Organoids overview bright-field pictures were taken with wide-field microscope (Mi8; Leica). For live imaging of lens formation (Video S2, **Figure 4**), organoids generated from *Foxe3::GFP* reporter line were imaged from day 1 (24 h post aggregation) to day 2 (40 h post aggregation). Imaging was caried out at 26°C directly in low binding 96-well plate (Thermo, Cat#:174925 Nunclon Sphera) with ACQUIFER Imaging Machine (ACQUIFER Imaging GmbH, Heidelberg, Germany) (Pandey et al., 2019). For each well, a set of 10 z-slices (70 µm step size) was acquired in the bright field (50% LED intensity, 40 ms exposure time) and 470 nm fluorescence (50% LED excitation source, FITC channel, 200 ms exposure time) channels. Imaging was performed over 16 hr of organoid development with 30 min intervals.

Transcriptome analysis

RNA was isolated from 1 dpf, 2 dpf, 3 dpf and 4 dpf medaka embryos as indicated in **Figure 7a**. Material from 6 embryos per condition was pooled. For embryos 1 and 2 dpf, head region containing optic vesicle and head structure anterior to the optic vesicles and eye cup and tissue anterior to the eye respectively were dissected in ice cold PBS. For stages 3 dpf and 4 dpf, eyes were dissected in ice cold PBS. Considering 3 h delay in organoid development with the respect to embryonic development (Zilova et al., 2021), organoids were collected for RNA isolation 3 h later than the embryonic tissue. For organoids, 20 organoids were pooled per condition. Samples were prepared in three replicates. Samples were rinsed with ice cold PBS and homogenized in 200 µl of Trizole and RNA was isolated using Direct-zol RNA Microprep (Zymoresearch, Cat#:R2060) according to the manufacture protocol. Library preparation was performed with the NEBNext Ultra II RNA Library Prep Kit for Illumina with NEBNext Poly A Selection Kit and NEBNext Multiplex Oligos with UMI. Input RNA was quantified with the Qubit BR RNA assay, and qualified on the Agilent Tapestation (RNA Assay), libraries were quantified with the Qubit DNA HS Assay, and qualified on the Agilent Tapestion (D1000 Assay). Libraries were sequenced on the Illumina NextSeq 550 Sequencer.

For downstream bioinformatic analysis, UMIs were extracted using UMI-Tools v1.1.2 (Smith et al., 2017). The first 12 bases of each read were trimmed, and reads shorter than 50 bases with an average quality lower than 20 were discarded using Trimmomatic v0.39 (Bolger et al., 2014). Processed reads were mapped to the Japanese medaka HdrR reference genome ASM223467v1 using Bowtie 2 v2.4.4 (Langmead and Salzberg, 2012), and deduplication of the mapped reads was performed with UMI-Tools v1.1.2 (Smith et al., 2017). Gene counts were extracted with featureCounts v2.0.3 (Liao et al., 2014) using the Ensembl version 113 GTF as a reference. The subsequent analysis was performed using R v.4.4.1 (*https://www.r-project.org/*). Genes with fewer than 10 counts across samples were excluded, and DESeq2 (Love et al., 2014) was used for normalization and variance stabilizing transformation of the gene counts. The transformed counts were then used for principal component analysis and heatmap visualization of lens-specific gene expression.

Generation of *FoxE3::GFP* reporter line

1.5 kb region upstream of *Foxe3* (ENSORLG00000017669) ATG was PCR amplified from medaka fish genomic DNA with Q5 High Fidelity DNA Polymerase (NEB, Cat#:M0491) using following primers: KpnI-FoxE3 (forward primer): 5'-TAAGGTACCTGGAATCAAGCAGAAATAATCCACCAAAAGCA and FoxE3_R1 (reverse primer): 5'-GGGGTACCGTTGCACATGCCAGAGAAGTAG, sub-cloned into pJet1.2/blunt cloning vector (Thermo Fisher Scientific), released via restriction digest with XhoI (NEB, Cat#:R0146) and NcoI (NEB, Cat#:R0193) restriction enzymes. The plasmid ISceI-pBSII KS+ (NovoPro, Cat#:V009906) (containing GFP sequence inserted via EcoRI and XbaI restriction sites) was digested with XhoI and NcoI, and ligated with the 1.5 kb *Foxe3* upstream regulatory element to generate *I-Sce/FoxE3(1.5kb)::*GFP. Plasmid DNA was isolated with QuiAprep Spin Miniprep Kit (Quiagen, Cat#:27106), sequenced (Eurofins Genomics) and analyzed using Geneious R8.1 (Biomatters). *I-Sce/FoxE3(1.5kb)::*GFP (10 ng/µl) was co-injected with Meganuclease (I-SceI) (NEB, Cat#:R0694L) into the cytoplasm of one cell stage medaka zygotes as previously described

(Thermes et al., 2002) to generate *Foxe3::GFP* reporter line. Embryos were raised and maintained at 26°C in 1× Embryo Rearing Medium (ERM, 17 mM NaCl, 40 mM KCl, 0.27 mM CaCl2, 0.66 mM MgSO4, 17 mM HEPES) and screened for ocular GFP expression 1 dpf on a Nikon SMZ18.

Quantitative analysis of organoid and lens size

Organoid and lens sizes (**Figure 5d** and **5f**) were measured as diameter (mean of 4 measurements per organoid) of the organoid on central optical section through organoid (organoid size at day 1) or on maximal projection of multiple z planes of the organoid (lens size at day 4). Depth of *Rx3::H2B-GFP* expression (**Figure 5e**) was measured on central optical section through the organoid (mean of 8 measurements per organoid) as a thickness of tissue carrying GFP^+ cells. Statistical analysis and plots were prepared with GraphPad. Figures were assembled with Affinity Designer 2.

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Additional information

Data Availability Statement

Raw data from bulk RNA sequencing (**Figure 7**) will be deposited upon acceptance in publicly available repository of Heidelberg University, HeiData.

Author Contributions Statement

Conceptualization: L.Z., J.W.

Investigation: E.S., M.A.D.T., L.Z., A.P., I.S.

Visualization: E.S., M.A.D.T., I.S., L.Z.

Funding acquisition: J.W., L.Z.

Project administration: J.W. Supervision: L.Z.

Writing – original draft: L.Z., J.W.

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Additional files

Supplemental PDF. *Figures S1-S3.*

Video S1. *Cells of the outer layer of the organoid acquire retina fate, related to Figure 3.*

Video S2. *Time laps imaging of multiple organoids from day 1 to day 2 showing the formation of the lens, related to Figure 4.*

Video S3. *Lens fate is acquired by cells localized in central part of the organoid, related to Figure 3.*

Video S4. *Arrangement of retinal and lens tissue in day 2 organoid, related to Figure 4.*

Data S1. *Excel table, related to transcriptome analysis of Figure 7. Normalized and variance stabilized counts labeled with Ensemble IDs.*

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Senior Editor
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University of Hong Kong, Hong Kong, Hong Kong

Reviewer #1 (Public review):

Summary:

The authors focused on medaka retinal organoids to investigate the mechanism underlying the eye cup morphogenesis. The authors succeeded to induce lens formation in fish retinal organoids using 3D suspension culture with minimal growth factor-containing media containing the Hepes. At day 1, Rx3:H2B-GFP＋ cells appear in the surface region of organoids. At day 1.5, Prox1+cells appear in the interface area between the organoid surface and the core of central cell mass, which develops a spherical-shaped lens later. So, Prox1+ cells covers the surface of the internal lens cell core. At day 2, foxe3:GFP+ cells appear in the Prox1+ area, where early lens fiber marker, LFC, starts to be expressed. In addition, foxe3:GFP+ cells show EdU+ incorporation, indicating that foxe3:GFP+ cells have lens epithelial cell-characters. At day 4, cry:EGFP+ cells differentiate inside the spherical lens core, whose surface area consists of LFC+ and Prox1+ cells. Furthermore, at day 4, the lens core moves towards the surface of retinal organoids to form an eyecup like structure, although this morphogenesis "inside out" mechanism is different from in vivo cellular "outside -in" mechanism of eye cup formation. From these data, the authors conclude that optic cup formation, especially the positioning of the lens, is established in retinal organoids though the different mechanism of in vivo morphogenesis.

Overall, manuscript presentation is nice. However, there are still obscure points to understand background mechanism. My comments are shown below.

Major comments:

(1) At the initial stage of retinal organoid morphogenesis, a spherical lens is centrally positioned inside the retinal organoids, by covering a central lens core by the outer cell sheet of retinal precursor cells. I wonder if the formation of this structure may be understood by differential cell adhesive activity or mechanical tension between lens core cells and retinal cell sheet, just like the previous study done by Heisenberg lab on the spatial patterning of endoderm, mesoderm and ectoderm (Nat. Cell Biol. 10, 429 - 436 (2008)). Lens core cells may be integrated inside retinal cell mass by cell sorting through the direct interaction between retinal cells and lens cells, or between lens cells and the culture media. After day 1, it is also possible to understand that lens core moves towards the surface of retinal organoids, if adhesive/tensile force states of lens core cells may be change by secretion of extracellular matrix. I wonder if the authors measure physical property, adhesive activity and solidness, of retinal precursor cells and lens core cells. If retinal organoids at day 1 are dissociated and cultured again, do they show the same patterning of internal lens core covering by the outer retinal cell sheet?

(2) Optic cup is evaginated from the lateral wall of neuroepithelium of the diencephalon. In zebrafish, cell movement occurs from the pigment epithelium to the neural retina during eye morphogenesis in an FGF-dependent manner. How the medaka optic cup morphogenesis is coordinated? I also wonder if the authors conduct the tracking of cell migration during optic cup morphogenesis to reveal how cell migration and cell division are regulated in lens of the Medaka retinal organoids. It is also interesting to examine how retinal cell movement is coordinated during Medaka retinal organoids.

(3) The authors showed that blockade of FGF signaling affects lens fiber differentiation in day 1-2, whereas lens formation seems to be intact in the presence of FGF receptor inhibitor in day 0-1. I suggest the authors to examine which tissue is a target of FGF signaling in retinal

organoids, using markers such as pea3, which is a downstream target of ERK branch of FGF signaling. Since FGF signaling promotes cell proliferation, is the lens core size normal in SU5402-treated organoids from day 0 to day 1?

(4) Fig. 3f and 3g indicate that there is some cell population located between foxe3:GFP+ cells and rx2:H2B-RFP+ cells. What kind of cell-type is occupied in the interface area between foxe3:GFP+ cells and rx2:H2B-RFP+ cells?

(5) Fig. 5e indicates the depth of Rx3 expression at day 1. Is the depth the thickness of Rx3 expressing cell sheet, which covers the central lens core in the organoids? If so, I wonder if total cell number of Rx3 expressing cell sheet may be different in each seeded-cell number, because thickness is the same across each seeded-cell number, but the surface area size may be different depending on underneath the lens core size. Please clarify this point.

(6) Noggin application inhibits lens formation at day 0-1. BMP signaling regulates formation of lens placode and olfactory placode at the early stage of development. It is interesting to examine whether Noggin-treated organoid expands olfactory placode area. Please check forebrain territory markers.

Significance:

Strength: This study is unique. The authors examined eye cup morphogenesis using fish retinal organoids. Eye cup normally consists of the lens, the neural retina, pigment epithelium and optic stalk. However, retinal organoids seem to be simple and consists of two cell types, lens and retina. Interestingly, a similar optic cup-like structure is achieved in both cases; however, underlying mechanism is different. It is interesting to investigate how eye morphogenesis is regulated in retinal organoids, under the unconstrained embryo-free environment.

Limitation: Description is OK, but analysis is not much profound. It is necessary to apply a bit more molecular and cellular level analysis, such as tracking of cell movement and visualization of FGF signaling in organoid tissues.

Advancement: The current study is descriptive. Need some conceptual advance, which impact cell biology field or medical science.

Audience: The target audience of current study are still within ophthalmology and neuroscience community people, maybe translational/clinical rather than basic biology. To beyond specific fields, need to formulate a general principle for cell and developmental biology.

https://doi.org/10.7554/eLife.108702.1.sa3

Reviewer #2 (Public review):

Summary:

In this study from Stahl et al., the authors demonstrate that medaka pluripotent embryonic cells can self-organise into eye organoids containing both retina and lens tissues. While these organoids can self-organize into an eye structure that resembles the vertebrate eye, they are built from a fundamentally different morphogenetic process - an "inside-out" mechanism where the lens forms centrally and moves outward, rather than the normal "outside-in" embryonic process. This is a very interesting discovery, both for our understanding of developmental biology and the potential for tissue engineering applications. The study would benefit from some additional experiments and a few clarifications. The authors suggest that the lens cells are the ones that move from the central to a more superficial position. Is this an active movement of lens cells or just the passive consequence of the retina cells acquiring a cup shape? Are the retina cells migrating behind the lens or the lens cells pushing outwards? High-resolution imaging of organoid cup formation, tracking retina cells in combination with membrane labeling of all cells would help elucidate the morphogenetic processes occurring in the organoids. Membrane labeling would also be useful as Prox1 positive lens cells appear elongated in embryos while in the organoids, cell shapes seem less organised, less compact and not elongated (for example as shown in Fig 3f,g).

The organoids could be a useful tool to address how cell fate is linked to cell shape acquisition. In the forming organoids, retinal tissue initially forms on the outside, while non-retinal tissue is located in the centre; this central tissue later expresses lens markers. Do the authors have any insights into why fate acquisition occurs in this pattern? Is there a difference in proliferation rates between the centrally located cells and the external ones? Could it be that highly proliferative cells give rise to neural retina (NR), while lower proliferating cells become lens?

What happens in organoids that do not form lenses? Do these organoids still generate foxe3 positive cells that fail to develop into a proper lens structure? And in the absence of lens formation, does the retina still acquire a cup shape?

The author suggest that lens formation occurs even in the absence of Matrigel. Is the process slower in these conditions? Are the resulting organoids smaller? While there are indeed some LFC expressing cells by day2, these cells are not very well organised and the pattern of expression seems dotty. Moreover, LFC staining seems to localise posterior to the LFC negative, lens-like structure (e.g. Fig.S1 3o'clock).

How do these organoids develop beyond day 4? Do they maintain their structural integrity at later stages?

The role of HEPES in promoting organoid formation is intriguing. Do the authors have any insights into why it is important in this context? Have the authors tried other culture conditions and does culture condition influence the morphogenetic pathways occurring within the organoids?

Significance:

This is a very interesting paper, and it will be important to determine whether this alternative morphogenetic process is specific to medaka or if similar developmental routes can be recapitulated in organoid cultures from other vertebrate species.

https://doi.org/10.7554/eLife.108702.1.sa2

Reviewer #3 (Public review):

Summary:

The manuscript by Stahl and colleagues reports an approach to generate ocular organoids composed of retinal and lens structures, derived from Medaka blastula cells. The authors present a comprehensive characterisation of the timeline followed by lens and retinal progenitors, showing these have distinct origins, and that they recapitulate the expression of differentiation markers found in vivo. Despite this molecular recapitulation, morphogenesis is strikingly different, with lens progenitors arising at the centre of the organoid, and subsequently translocating to the outside.

Major Comments:

- The manuscript presents a beautiful set of high-quality images showing expression of lens differentiation markers over time in the organoids. The set of experiments is very robust, with high numbers of organoids analysed and reproducible data. The mechanism by which lens specification is promoted in these organoids is, however, poorly analysed, and the reader does not get a clear understanding of what is different in these experiments, as compared to previous attempts, to support lens differentiation. There is a mention to HEPES supplementation, but no further analysis is provided, and the fact that the process is independent of ECM contradicts, as the authors point out, previous reports. The manuscript would benefit from a more detailed analysis of the mechanisms that lead to lens differentiation in this setting.

- The markers analysed to show onset of lens differentiation in the organoids seem to start being expressed, in vivo, when the lens placode starts invaginating. An analysis of earlier stages is not presented. This would be very informative, allowing to determine whether progenitors differentiate as placode and neuroepithelium first, to subsequently continue differentiating into lens and retina, respectively. Could early placodal and anterior neural plate markers be analysed in the organoids? This would provide a more complete sequence of lens vs retina differentiation in this model.

- The analysis of BMP and Fgf requirement for lens formation and differentiation is suggestive, but the source of these signals is not resolved or mentioned in the manuscript. Are BMP4 and Fgf8 expressed by the organoids? Where are they coming from?

- The fact that the lens becomes specified in the centre of the organoid is striking, but it is for me difficult to visualise how it ends up being extruded from the organoid. Did the authors try to follow this process in movies? I understand that this may be technically challenging, but it would certainly help to understand the process that leads to the final organisation of retinal and lens tissues in the organoid. There is no discussion of why the morphogenetic mechanism is so different from the in vivo situation. The manuscript would benefit from explicitly discussing this.

Significance:

This study describes a reproducible approach to differentiate ocular organoids composed of lens and retinal tissues. The characterisation of lens differentiation in this model is very detailed, and despite the morphogenetic differences, the molecular mechanisms show many similarities to the in vivo situation. The manuscript however does not highlight, in my opinion, why this model may be relevant. Clearly articulating this relevance, particularly in the discussion, will enhance the study and provide more clarity to the readers regarding the significance of the study for the field of organoid research, ocular research and regenerative studies.

https://doi.org/10.7554/eLife.108702.1.sa1

Author response:

Description of the planned revisions

> ***Reviewer #1 (Evidence, reproducibility and clarity):***
>
> *Summary*
>
> *The authors focused on medaka retinal organoids to investigate the mechanism underlying the eye cup morphogenesis. The authors succeeded to induce lens formation in fish retinal organoids using 3D suspension culture with minimal growth factor-*

> *containing media containing the Hepes. At day 1, Rx3:H2B-GFP＋cells appear in the surface region of organoids. At day 1.5, Prox1+cells appear in the interface area between the organoid surface and the core of central cell mass, which develops a spherical-shaped lens later. So, Prox1+ cells covers the surface of the internal lens cell core. At day 2, foxe3:GFP+ cells appear in the Prox1+ area, where early lens fiber marker, LFC, starts to be expressed. In addition, foxe3:GFP+ cells show EdU+ incorporation, indicating that foxe3:GFP+ cells have lens epithelial cell-characters. At day 4, cry:EGFP+ cells differentiate inside the spherical lens core, whose the surface area consists of LFC+ and Prox1+ cells. Furthermore, at day 4, the lens core moves towards the surface of retinal organoids to form an eye-cup like structure, although this morphogenesis "inside out" mechanism is different from in vivo cellular "outside -in" mechanism of eye cup formation. From these data, the authors conclude that optic cup formation, especially the positioning of the lens, is established in retinal organoids though the different mechanism of in vivo morphogenesis.*
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> *(1) At the initial stage of retinal organoid morphogenesis, a spherical lens is centrally positioned inside the retinal organoids, by covering a central lens core by the outer cell sheet of retinal precursor cells. I wonder if the formation of this structure may be understood by differential cell adhesive activity or mechanical tension between lens core cells and retinal cell sheet, just like the previous study done by Heisenberg lab on the spatial patterning of endoderm, mesoderm and ectoderm (Nat. Cell Biol. 10, 429 - 436 (2008)). Lens core cells may be integrated inside retinal cell mass by cell sorting through the direct interaction between retinal cells and lens cells, or between lens cells and the culture media. After day 1, it is also possible to understand that lens core moves towards the surface of retinal organoids, if adhesive/tensile force states of lens core cells may be change by secretion of extracellular matrix. I wonder if the authors measure physical property, adhesive activity and solidness, of retinal precursor cells and lens core cells. If retinal organoids at day 1 are dissociated and cultured again, do they show the same patterning of internal lens core covering by the outer retinal cell sheet?*

The question, whether different adhesive activity is involved in cell sorting and lens formation is indeed very intriguing. To address this point, we will include additional experiment (see Revision Plan, experiment 1). This experiment will be based on the dissociation and re-aggregation of lens-forming organoids as suggested by the reviewer. To monitor cell type specific sorting, we will employ a lens progenitor reporter line Foxe3::GFP and the retina-specific Rx2::H2B-RFP. If different adhesive activities of lens and retinal progenitor cells are involved and drive the process of cell sorting, dissociation and re-aggregation will result in cell sorting based on their identity.

> *(2) Optic cup is evaginated from the lateral wall of neuroepithelium of the diencephalon. In zebrafish, cell movement occurs from the pigment epithelium to the neural retina during eye morphogenesis in an FGF-dependent manner. How the medaka optic cup morphogenesis is coordinated? I also wonder if the authors conduct the tracking of cell migration during optic cup morphogenesis to reveal how cell migration and cell division are regulated in lens of the Medaka retinal organoids. It is also interesting to examine how retinal cell movement is coordinated during Medaka retinal organoids.*

Looking into the detail of how optic cup-looking tissue arrangement of ocular organoids is achieved on cellular level is of course interesting. Our previous study showed that optic vesicles of medaka retinal organoids do not form optic cups (for details please see Zilova et al., 2021, eLIFE). We assume that the formation of cup-looking structure of the ocular organoids is mediated by the following processes: establishment of retina and lens domains at the specific region of the organoid – retina on the surface and lens in the center (see Figure S2 d and Figure 3e, and Figure 4). Further dislocation of the centrally formed lens towards the organoid periphery through the retina layer, places the lens to the periphery while retinal cells stay static. We assume that the "cup-like" shape is acquired by extrusion of the lens from the center of the organoid. To clarify this process with respect to tissue rearrangements and cell movements, we will include additional experiments (see Revision Plan, experiment 2) and follow lens- and retina-fated cells (by employing lens-specific Foxe3::GFP and retina-specific Rx2::H2B-RFP reporter lines) through the process of lens extrusion to dissect individual contribution of retinal/lens cells to this process (cross-reference with Reviewer #2).

> *(3) The authors showed that blockade of FGF signaling affects lens fiber differentiation in day 1-2, whereas lens formation seems to be intact in the presence of FGF receptor inhibitor in day 0-1. I suggest the authors to examine which tissue is a target of FGF signaling in retinal organoids, using markers such as pea3, which is a downstream target of ERK branch of FGF signaling. Since FGF signaling promotes cell proliferation, is the lens core size normal in SU5402-treated organoids from day 0 to day 1?*

Assessing the activity of FGF signaling (cross-reference to Reviewer #3) in the organoids is indeed an important point. To address which tissue is the target of FGF signaling we will include additional experiments and assess the phosphorylation status of ERK (pERK) and expression of the ERK downstream target pea3, as suggested by the reviewer (see Revision Plan, experiment 3). That will allow to identify the tissue within the organoid responding to the Fgf signaling.

Lens core size of organoids treated with SU5402 from day 0 to day 1 is fully comparable to the control (please see Figure 6b).

> *(4) Fig. 3f and 3g indicate that there is some cell population located between foxe3:GFP+ cells and rx2:H2B-RFP+ cells. What kind of cell-type is occupied in the interface area between foxe3:GFP+ cells and rx2:H2B-RFP+ cells?*

That is for sure an interesting question. We are aware of this population of cells. We currently do not have data that would with certainty clarify the fate of those cells. We are currently following up on that question with the use of scRNA sequencing, however we will not be able to address this question in the current manuscript.

> *(5) Fig. 5e indicates the depth of Rx3 expression at day 1. Is the depth the thickness of Rx3 expressing cell sheet, which covers the central lens core in the organoids? If so, I wonder if total cell number of Rx3 expressing cell sheet may be different in each seeded-cell number, because thickness is the same across each seeded-cell number, but the surface area size may be different depending on underneath the lens core size. Please clarify this point.*

Yes. Figure 5e indicates the thickness of the cell sheet expressing Rx3 that lies on the surface of the organoid. Indeed, the number of Rx3-expressing cells (and lens cells) scales with the size of the organoid as stated in the submitted manuscript.

> *(6) Noggin application inhibits lens formation at day 0-1. BMP signaling regulates formation of lens placode and olfactory placode at the early stage of development. It is interesting to examine whether Noggin-treated organoid expands olfactory placode area. Please check forebrain territory markers.*

What tissue differentiates at the expense of the lens in BMP inhibitor-treated organoids is of course an intriguing question. To address the identity of cells differentiated under this condition we will include an additional experiment (see Revision Plan, experiment 4 as suggested by the reviewer). We will check for the expression of Lhx2, Otx2 and Huc/D to address this point.

> *I have no minor comments*
>
> ***Referees cross-commenting***
>
> *I agree that all reviewers have similar suggestions, which are reasonable and provided the same estimated time for revision.*
>
> ***Reviewer #1 (Significance):***
>
> *Strength:*
>
> *This study is unique. The authors examined eye cup morphogenesis using fish retinal organoids. Eye cup normally consists of the lens, the neural retina, pigment epithelium and optic stalk. However, retinal organoids seem to be simple and consists of two cell types, lens and retina. Interestingly, a similar optic cup-like structure is achieved in both cases; however, underlying mechanism is different. It is interesting to investigate how eye morphogenesis is regulated in retinal organoids,under the unconstrained embryo-free environment.*
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> *Limitation:*
>
> *Description is OK, but analysis is not much profound. It is necessary to apply a bit more molecular and cellular level analysis, such as tracking of cell movement and visualization of FGF signnaling in organoid tissues.*
>
> *Advancement:*
>
> *The current study is descriptive. Need some conceptual advance, which impact cell biology field or medical science.*
>
> *Audience:*
>
> *The target audience of current study are still within ophthalmology and neuroscience community people, maybe translational/clinical rather than basic biology. To beyond specific fields, need to formulate a general principle for cell and developmental biology.*
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> ***Reviewer #2 (Evidence, reproducibility and clarity):***
>
> *In this study from Stahl et al., the authors demonstrate that medaka pluripotent embryonic cells can self-organise into eye organoids containing both retina and lens tissues. While these organoids can self-organize into an eye structure that resembles the vertebrate eye, they are built from a fundamentally different morphogenetic process – an "inside-out" mechanism where the lens forms centrally and moves outward, rather than the normal "outside-in" embryonic process. This is a very interesting discovery, both for our understanding of developmental biology and the potential for tissue engineering applications. The study would benefit from some additional experiments and a few clarifications.*
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> *The authors suggest that the lens cells are the ones that move from the central to a more superficial position. Is this an active movement of lens cells or just the passive consequence of the retina cells acquiring a cup shape? Are the retina cells migrating behind the lens or the lens cells pushing outwards? High-resolution imaging of organoid*

> *cup formation, tracking retina cells in combination with membrane labeling of all cells would help elucidate the morphogenetic processes occurring in the organoids. Membrane labeling would also be useful as Prox1 positive lens cells appear elongated in embryos while in the organoids, cell shapes seem less organised, less compact and not elongated (for example as shown in Fig 3f,g).*

Looking into the detail of how optic cup-looking tissue arrangement of ocular organoids is achieved on cellular level is of course interesting. We assume that the formation of cup-looking structures of the ocular organoids is mediated by following processes: establishment of retina and lens domains at a specific region of the organoid – retina on the surface and lens in the center (see Figure S2 d and Figure 3e, and Figure 4). Further dislocation of centrally formed lenses towards the organoid periphery through the retina layer, place the lens to the periphery while retinal cells stay static. We assume that the "cup-like" shape is acquired by extrusion of the lens. To clarify this process with respect to tissue rearrangements and cell movements, we will include additional experiments (see Revision Plan, experiment 2). We will follow lens- and retina-fated cells (by employing lens-specific Foxe3::GFP and retina-specific Rx2::H2B-RFP reporter lines) through the process of lens extrusion to dissect the individual contribution of retinal/lens cells to this process (cross-reference with Reviewer #1).

> *The organoids could be a useful tool to address how cell fate is linked to cell shape acquisition. In the forming organoids, retinal tissue initially forms on the outside, while non-retinal tissue is located in the centre; this central tissue later expresses lens markers. Do the authors have any insights into why fate acquisition occurs in this pattern? Is there a difference in proliferation rates between the centrally located cells and the external ones? Could it be that highly proliferative cells give rise to neural retina (NR), while lower proliferating cells become lens?*

The question how is the retinal and lens domain established in this specific manner is indeed intriguing and very interesting. We dedicated a part of the discussion to this topic. We discuss the role of the diffusion limit and the potential contribution of BMB and FGF signaling to this arrangement. Additional experiments (see Revision Plan, experiment 3) addressing the source and target tissues of FGF and BMP signaling in the organoid will ultimately bring more clarity to our understanding of the tissue arrangements in the organoid.

Although analysis of the proliferation rate of the cells at the surface and in the central region of the organoid might possibly show some differences in the proliferation rates between lens and retinal cells, we do not have any indications, that the proliferation rate itself would be instructive or superior to the cell fate decisions.

> *What happens in organoids that do not form lenses? Do these organoids still generate foxe3 positive cells that fail to develop into a proper lens structure? And in the absence of lens formation, does the retina still acquire a cup shape?*

Lens formation is primarily dependent on acquisition/specification of Foxe3-expressing lens placode progenitors. If those are not present, a lens does not develop. Once Foxe3-expressing progenitors are established, a lens is formed in unperturbed conditions (measured by the presence of expression of crystallin proteins). In such conditions, organoids that do not have a lens, do not carry Foxe3-expressing cells.

In the absence of the lens, the organoid is composed of retinal neuroepithelium, that does not form an optic cup (for details of such phenotypes please see Zilova et al., 2021, eLIFE).

> *The author suggest that lens formation occurs even in the absence of Matrigel. Is the process slower in these conditions? Are the resulting organoids smaller? While there are*

indeed some LFC expressing cells by day2, these cells are not very well organised and the pattern of expression seems dotty. Moreover, LFC staining seems to localise posterior to the LFC negative, lens-like structure (e.g. Fig.S1 3o'clock).

How do these organoids develop beyond day 4? Do they maintain their structural integrity at later stages?

The role of HEPES in promoting organoid formation is intriguing. Do the authors have any insights into why it is important in this context? Have the authors tried other culture conditions and does culture condition influence the morphogenetic pathways occurring within the organoids?

We thank the reviewer for pointing this out. We were not clear in the wording and describing of our observation. Indeed, Matrigel is not required for acquisition of lens fate, which can be demonstrated with the expression of lens-specific markers. However, the presence of Matrigel has a profound impact on the structural aspects of organoid formation. Matrigel is essential for organization of retinal-committed cells into the retinal epithelium (Zilova et al., 2021, eLIFE). The absence of the structure of the retinal epithelium can indeed negatively impact on the cellular organization and the overall lens structure. To clarify the contribution of the Matrigel to the speed of organoid lens development and to the overall structure of the organoid lens we will perform additional experiments (see Revision Plan, experiment 5). With the use of Foxe3::GFP reporter line we will measure the onset of the lens-specific gene expression. In addition, we will use the immunohistochemistry to assess the gross morphology and size of the organoids grown without the Matrigel (cross-reference with Reviewer #3).

The role of the HEPES in lens formation is indeed very intriguing and currently under investigation. As HEPES is mainly used to regulate pH of the culture media and pH might have an impact on multiple cellular processes, it will require significant time investment to dissect molecular mechanism underlying the effect of HEPES on the process of lens formation (cross reference with Reviewer #3) and therefore cannot be addressed in the current manuscript.

Referees cross-commenting

Pleased to see that all the other reviewers are positive about the study and raise similar concerns and comments

Reviewer #2 (Significance):

This is a very interesting paper, and it will be important to determine whether this alternative morphogenetic process is specific to medaka or if similar developmental routes can be recapitulated in organoid cultures from other vertebrate species.

Reviewer #3 (Evidence, reproducibility and clarity):

Summary:

The manuscript by Stahl and colleagues reports an approach to generate ocular organoids composed of retinal and lens structures, derived from Medaka blastula cells. The authors present a comprehensive characterisation of the timeline followed by lens and retinal progenitors, showing these have distinct origins, and that they recapitulate the expression of differentiation markers found in vivo. Despite this molecular recapitulation, morphogenesis is strikingly different, with lens progenitors arising at the centre of the organoid, and subsequently translocating to the outside.

Comments:

> *- The manuscript presents a beautiful set of high quality images showing expression of lens differentiation markers over time in the organoids. The set of experiments is very robust, with high numbers of organoids analysed and reproducible data. The mechanism by which lens specification is promoted in these organoids is, however, poorly analysed, and the reader does not get a clear understanding of what is different in these experiments, as compared to previous attempts, to support lens differentiation. There is a mention to HEPES supplementation, but no further analysis is provided, and the fact that the process is independent of ECM contradicts, as the authors point out, previous reports. The manuscript would benefit from a more detailed analysis of the mechanisms that lead to lens differentiation in this setting.*

The role of the HEPES in lens formation is indeed very intriguing and under current investigation. As HEPES is mainly used to regulate pH of the culture media and pH might have an impact on multiple cellular processes it will require a significant time investment to dissect molecular mechanism underlying the effect of HEPES on the process of lens formation (cross reference with Reviewer #2) and therefore unfortunately cannot be addressed in the current manuscript.

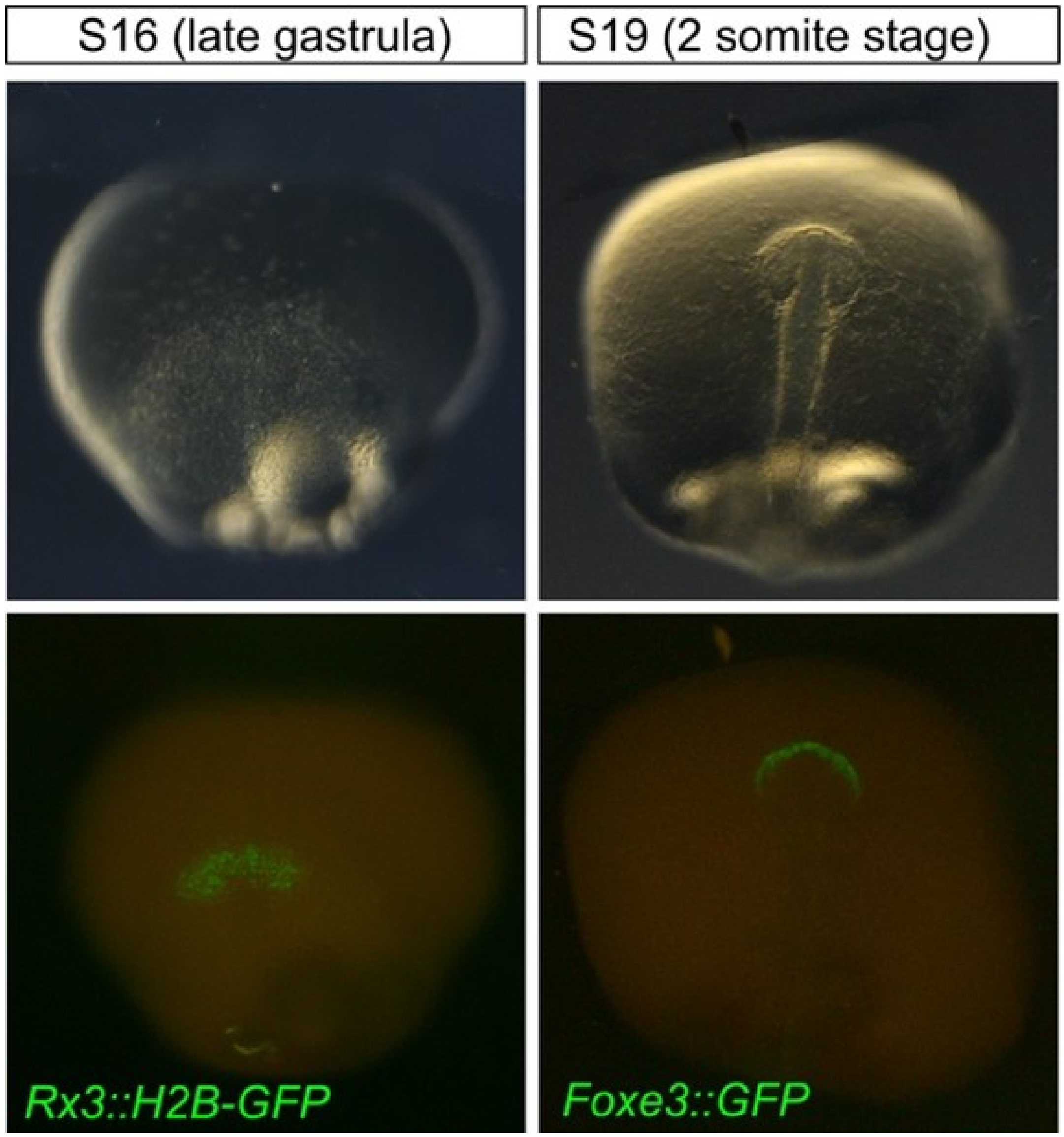
To clarify the contribution of the Matrigel to the organoid lens development we will perform additional experiments (see Revision Plan, experiment 5). With the use of Foxe3::GFP reporter line we will measure the onset of the lens-specific gene expression. In addition, we will use the immunohistochemistry to assess the gross morphology and size of the organoids grown without the Matrigel (cross-reference with Reviewer #2).

> *- The markers analysed to show onset of lens differentiation in the organoids seem to start being expressed, in vivo, when the lens placode starts invaginating. An analysis of earlier stages is not presented. This would be very informative, allowing to determine whether progenitors differentiate as placode and neuroepithelium first, to subsequently continue differentiating into lens and retina, respectively. Could early placodal and anterior neural plate markers be analysed in the organoids? This would provide a more complete sequence of lens vs retina differentiation in this model.*

Yes. The figures show the expression of lens and retinal markers in the embryo in later developmental stages and the timing of their expression can be documented with higher temporal resolution. In the revised version of the manuscript, we will provide the information about the onset of expression of Rx3::H2B-GFP (retina) and Foxe3::GFP (lens) (see Author response image 1). Rx3 represents one of the earlies markers labeling the presumptive eye field within the region of the anterior neural plate (S16, late gastrula). FoxE3::GFP expression can be detected within the head surface ectoderm before the lens placode is formed showing that Foxe3 is a suitable marker of placodal progenitors in medaka.

We are convinced that the onset of Rx3 and Foxe3-driven reporters is early enough to make the claim about the separate origin of the lens (placodal) and retinal (anterior neuroectoderm) tissues within the ocular organoids.

Author response image 1.



> *- The analysis of BMP and Fgf requirement for lens formation and differentiation is suggestive, but the source of these signals is not resolved or mentioned in the manuscript. Are BMP4 and Fgf8 expressed by the organoids? Where are they coming from?*

Indeed, addressing the source of BMP and FGF activation would bring more clarity in understanding the mechanism of retina/lens specification within the ocular organoids (cross reference with Reviewer #1). To address this point, we will include additional experiments (see Revision Plan, experiment 3). We will analyze the expression of respective ligands (Bmp4 and Fgf8) and activation of downstream effectors of BMP and FGF signaling pathways within the ocular organoids as suggested by Reviewer #1 and Reviewer #3.

> *- The fact that the lens becomes specified in the centre of the organoid is striking, but it is for me difficult to visualise how it ends up being extruded from the organoid. Did the authors try to follow this process in movies? I understand that this may be technically*

> *challenging, but it would certainly help to understand the process that leads to the final organisation of retinal and lens tissues in the organoid. There is no discussion of why the morphogenetic mechanism is so different from the in vivo situation. The manuscript would benefit from explicitly discussing this.*

Following the extruding lens in vivo is indeed very relevant suggestion. To clarify the process of ocular organoid formation in the respect of tissue rearrangements and cell movements, we will include additional experiment (see Revision Plan, experiment 2). We will follow lens- and retina-fated cells (by employing lens-specific Foxe3::GFP and retina-specific Rx2::H2B-RFP reporter lines) through the process of lens extrusion (cross-reference with Reviewer #1 and Reviewer #2).

> ***Referees cross-commenting***
>
> *We all seem to have similar comments and concerns. I think overall the suggestions are feasible and realistic for the timeframe provided.*
>
> ***Reviewer #3 (Significance):***
>
> *This study describes a reproducible approach to differentiate ocular organoids composed of lens and retinal tissues. The characterisation of lens differentiation in this model is very detailed, and despite the morphogenetic differences, the molecular mechanisms show many similarities to the in vivo situation. The manuscript however does not highlight, in my opinion, why this model may be relevant. Clearly articulating this relevance, particularly in the discussion, will enhance the study and provide more clarity to the readers regarding the significance of the study for the field of organoid research, ocular research and regenerative studies.*

Revision Plan:

(1) To address whether differential adhesion properties of retinal and lens progenitors mediate cell sorting to establish retina and lens domains in the organoids (Reviewer #1, comment 1), we will perform dissociation of the organoids on day 1 and subsequential re-aggregation. This experiment will allow to follow cell type specific adhesion properties of lens and retinal progenitor cells. We will employ lens progenitor reporter line Foxe3::GFP and retina-specific Rx2::H2B-RFP to monitor cell type specific sorting with fluorescent microscopy.

(2) Multiple reviewers (Reviewer #1, Reviewer #2, Reviewer #3) asked for the presentation of detailed in vivo imaging experiment showing individual contributions of retina- and lens-fated cells to the resulting tissue organization withing the ocular organoid. We will perform in vivo live imaging experiment to follow the movements of individual lens (Foxe3::GFP) and retinal (Rx2::H2B-GFP) cells from day 1 to day 2 of organoid development to address this point.

(3) Reviewer #1 and Reviewer #3 raised questions concerning the role of FGF and BMP signaling and sources of these signaling pathway activities in ocular organoid tissue arrangement. To address this point and bring more light into the molecular mechanisms regulating lens and retina tissue arrangement in the organoid, we will perform additional experiment. We will assess the expression of candidate FGF and BMP ligands (Fgf8, Bmp7 and Bmp4) and activation of downstream effectors (p-ERK, p-SMAD) and the direct transcriptional target of Fgf signaling (Pea3) in the developing organoids. This will allow the identification of the tissue producing the ligand on one site and tissue responding to the signaling on the other site and help out to narrow down the molecular mechanism controlling tissue arrangements in the organoid.

(4) We will analyze the expression of forebrain territory markers in organoids treated with the BMP inhibitor to identify the identity of the tissue differentiating at the expense of lens under the BMP inhibition (suggested by Reviewer #1). We will label Noggin-treated organoids with the antibodies against Lhx2, Otx2 and HuC/D to address this point.

(5) We will provide more comprehensive analysis of the organoids grown without the Matrigel and compare them to the organoids grown in the presence of the Matrigel (mentioned by Reviewer #2 and Reviewer #3). With the use of lens progenitor-specific Foxe3::GFP reporter line, we will measure the onset of the lens-specific gene expression. In addition, we will use the immunohistochemistry to assess the gross morphology and size of the organoids grown without the Matrigel.

Description of analyses that authors prefer not to carry out

> ***Reviewer #1:***
>
> *(4) Fig. 3f and 3g indicate that there is some cell population located between foxe3:GFP+ cells and rx2:H2B-RFP+ cells. What kind of cell-type is occupied in the interface area between foxe3:GFP+ cells and rx2:H2B-RFP+ cells?*

That is for sure interesting question. We are aware of this population of cells. We currently do not have a data that would with certainty clarify the fate of those cells. We are currently following up on that question with the use of scRNA sequencing, however we will not be able to address this question in the current manuscript.

> ***Reviewer #2:***
>
> *The role of HEPES in promoting organoid formation is intriguing. Do the authors have any insights into why it is important in this context? Have the authors tried other culture conditions and does culture condition influence the morphogenetic pathways occurring within the organoids?*

The role of the HEPES in lens formation is indeed very intriguing and under current investigation. As HEPES is mainly used to regulate pH of the culture media and pH might have impact on multiple cellular processes it will require significant time investment to dissect molecular mechanism underlying the effect of the HEPES on the process of lens formation (cross reference with Reviewer #3) and cannot be addressed in the current manuscript.

> *Is there a difference in proliferation rates between the centrally located cells and the external ones? Could it be that highly proliferative cells give rise to neural retina (NR), while lower proliferating cells become lens?*

Although analysis of the proliferation rate of the cells at the surface and in the central region of the organoid might possibly show some differences in the proliferation rates between lens and retinal cells, we do not have any indications, that the proliferation rate itself would be instructive or superior to the cell fate decisions.

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